
Magnetically Coupled Circuits

Words learning

Disciplines applying EM

- electric machines,
- electromechanical energy conversion,
- radar meteorology,
- remote sensing,
- satellite communications,
- bioelectromagnetics,
- electromagnetic interference and compatibility,
- plasmas,
- fiber optics

EM devices

- electric motors and generators,
- transformers,
- autotransformers,
- electromagnets,
- magnetic levitation,
- antennas,
- radars,
- microwave ovens,
- microwave dishes,
- Superconductors
- electrocardiograms.

Outline

□ Mutual Inductance

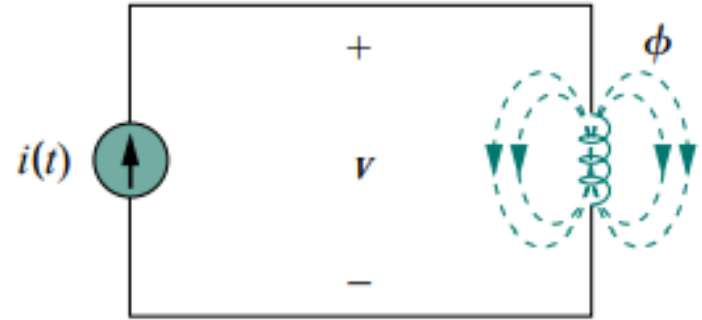
□ Analysis of Circuits with Mutual Inductance

□ The Transformer

Mutual Inductance

Recap of Self Inductance

1. According to Faraday's law, the voltage v induced in the coil is proportional to the number of turns N and the time rate of change of the magnetic flux ϕ .



$$v(t) = \frac{d\psi}{dt} = N \frac{d\phi}{dt}$$

2. The flux produced by a current is proportional to the current.

$$\phi = \int_S \mathbf{B} \cdot d\mathbf{S} \propto i$$

(For long straight helix tube, $B = \mu \frac{N}{l} i = \mu n i$)

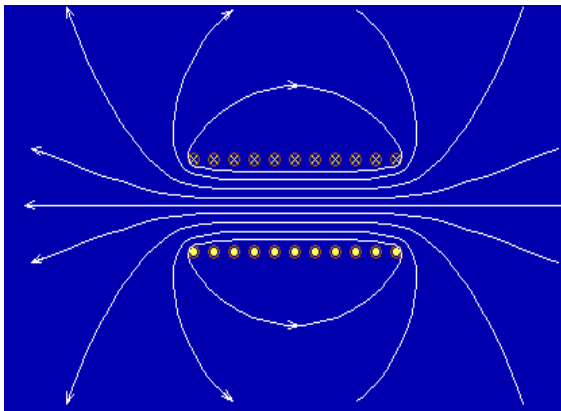
Recap of Self Inductance

3. Self Inductance

$$v(t) = N \frac{d\phi}{dt} = N \frac{d\phi}{di} \frac{di}{dt} = L \frac{di}{dt}$$

$$L = N \frac{d\phi}{di} \quad (\Psi = Li)$$

4. Self Inductance depends on the permeability μ , the turns N and the size (l, S) of the coil.



$$\Psi = NBS = nl \cdot \mu ni \cdot S$$

$$L = \frac{\Psi}{i} = \mu n^2 Sl = \mu n^2 V$$

Mutual Inductance

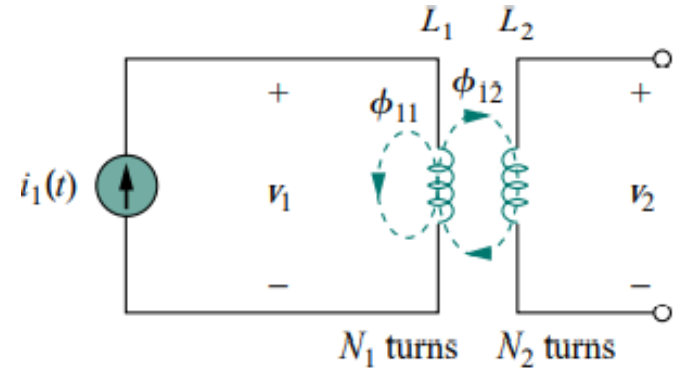
L_1 and L_2 are in close proximity with each other and one of them is open circuited

(a) A current i_1 at L_1 produces B_1 ,

B_1 links only L_1 — ϕ_{11} ,

B_1 links both L_1 and L_2 — ϕ_{12}

The flux in coil 1 — $\phi_1 = \phi_{11} + \phi_{12}$



L_2 is open circuited

$$v_1(t) = N_1 \frac{d\phi_1}{dt} = N_1 \frac{d\phi_1}{di_1} \frac{di_1}{dt} = L_1 \frac{di_1}{dt} \quad \text{— self inductance in coil 1}$$

$$v_2(t) = N_2 \frac{d\phi_{12}}{dt} = N_2 \frac{d\phi_{12}}{di_1} \frac{di_1}{dt} = M_{21} \frac{di_1}{dt}$$

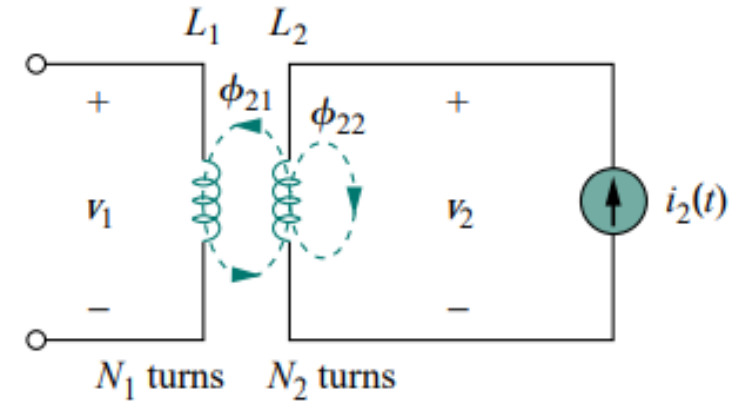
M_{21} — mutual inductance of coil 2 with respect to coil 1

Mutual Inductance

(b) A current i_2 at L_2 produces B_2 ,

B_2 links only L_2 — ϕ_{22} ,

B_2 links both L_1 and L_2 — ϕ_{21}



L_1 is open circuited

The flux in coil 2 — $\phi_2 = \phi_{21} + \phi_{22}$

$$v_1(t) = N_1 \frac{d\phi_{21}}{dt} = N_1 \frac{d\phi_{21}}{di_2} \frac{di_2}{dt} = M_{12} \frac{di_2}{dt}$$

M_{12} — mutual inductance of coil 1 with respect to coil 2

$$v_2(t) = N_2 \frac{d\phi_2}{dt} = N_2 \frac{d\phi_2}{di_2} \frac{di_2}{dt} = L_2 \frac{di_2}{dt}$$

— self inductance in coil 2

(c) $M_{12} = M_{21} = M \text{ (H)}$

Mutual Inductance

Conclusion:

Mutual inductance is the ability of one inductor to induce a voltage across a neighboring inductor, measured in henrys (H)

- (1) The inductors or coils are in close proximity.
- (2) The circuits are driven by time-varying sources.
- (3) M is always a positive quantity.
- (4) The choice of the correct polarity for $M di/dt$ is made by examining the orientation or particular way in which both coils are physically wound and applying Lenz's law in conjunction with the right-hand rule.

Polarity of Mutual Voltage

Combined Mutual and Self-induced Voltage

(No terminal is open circuited)

(a) According to current directions and right-hand rule, we can decide the total flux in each inductor.

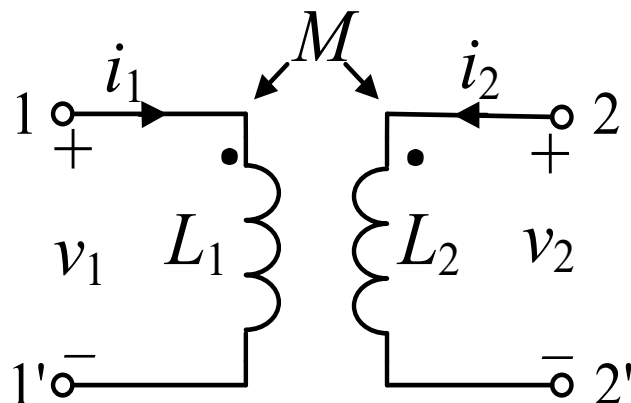
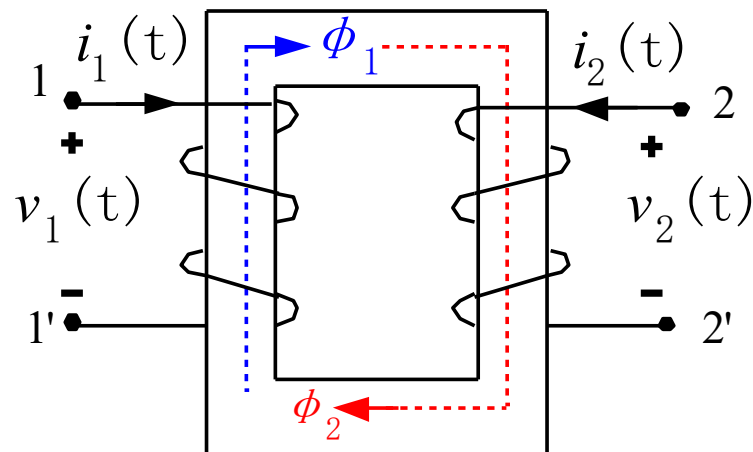
$$\psi_1(t) = L_1 i_1 + M i_2$$

$$\psi_2(t) = L_2 i_2 + M i_1$$

(b) According to current and voltage reference, we can decide the sign '+' or '-' in Faraday's law.

$$v_1 = \frac{d\psi_1}{dt} = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$v_2 = \frac{d\psi_2}{dt} = L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$

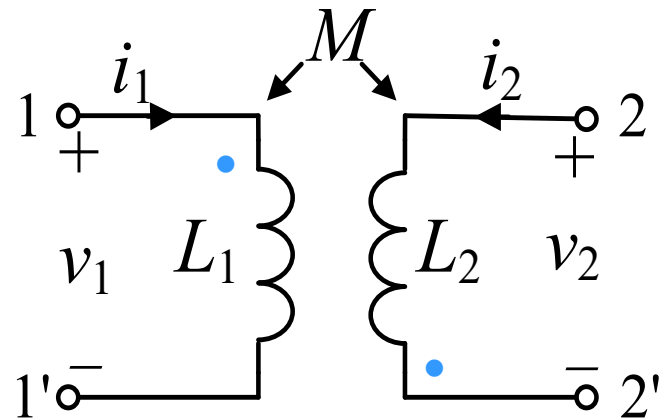
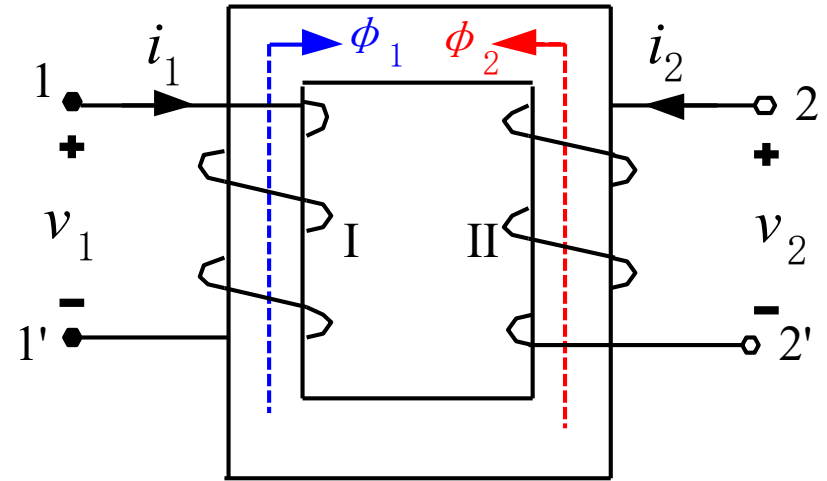


Polarity of Mutual Voltage

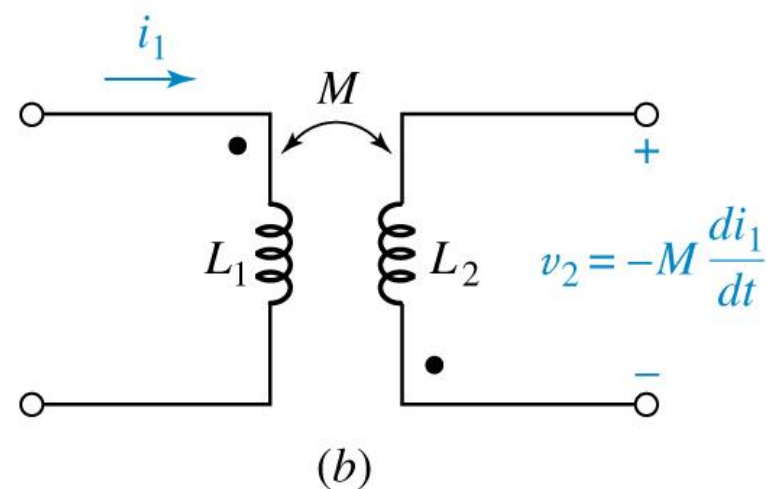
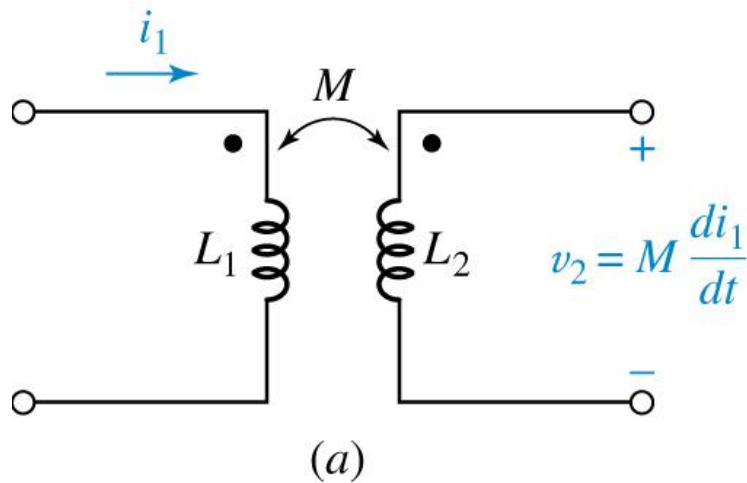
Combined Mutual and Self-induced Voltage

$$v_1 = L \frac{di_1}{dt} - M \frac{di_2}{dt}$$

$$v_2 = -M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$$

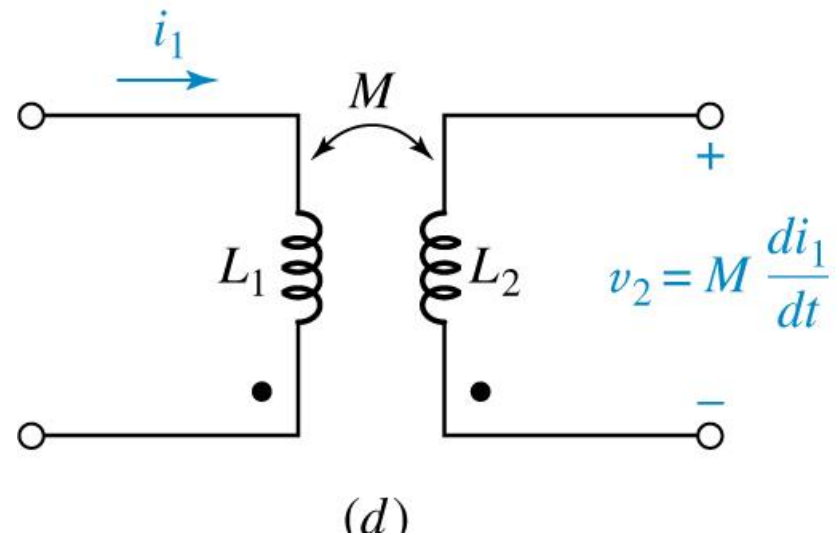
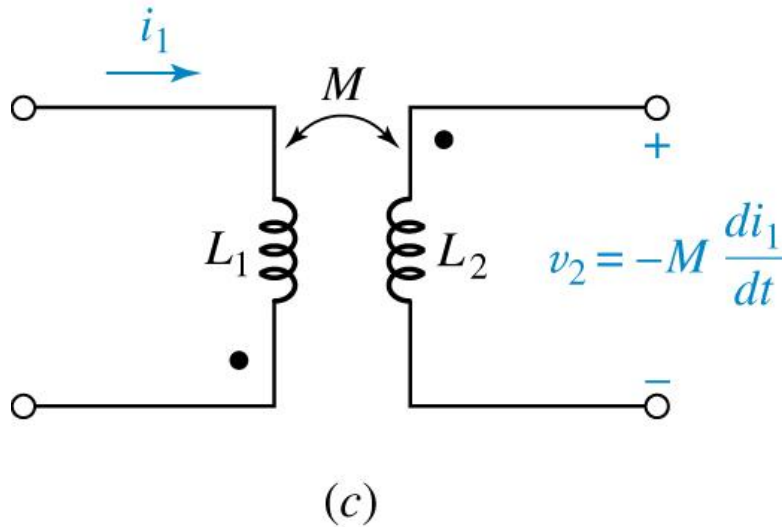


Dot Convention



If a current enters the dotted terminal of one coil, the reference polarity of the mutual voltage in the second coil is positive at the dotted terminal of the second coil.

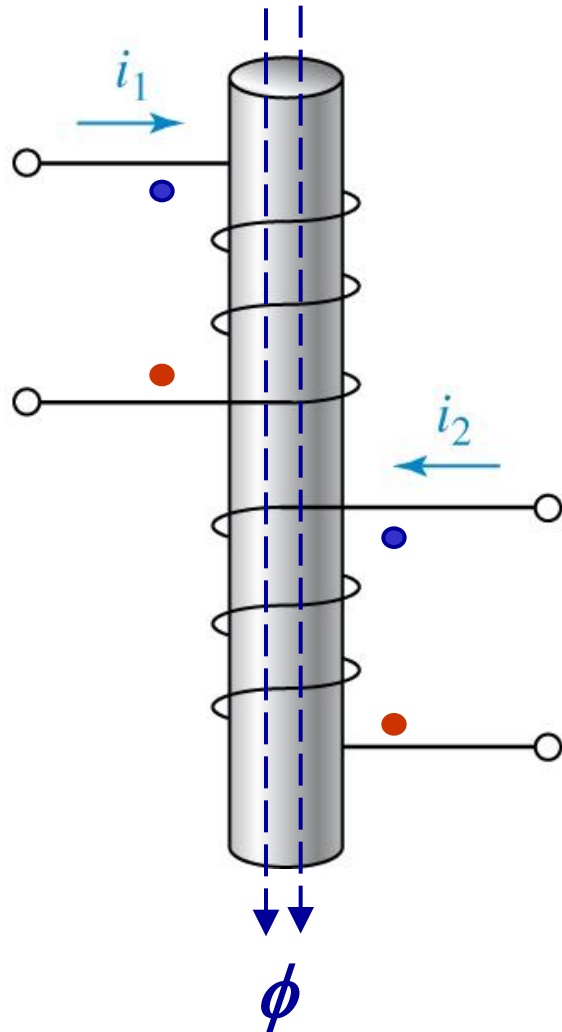
Dot Convention



Current entering the undotted terminal of one coil produces a voltage that is sensed positively at the undotted terminal of the second coil.

Dot Convention

Physical Basis of Dot Convention



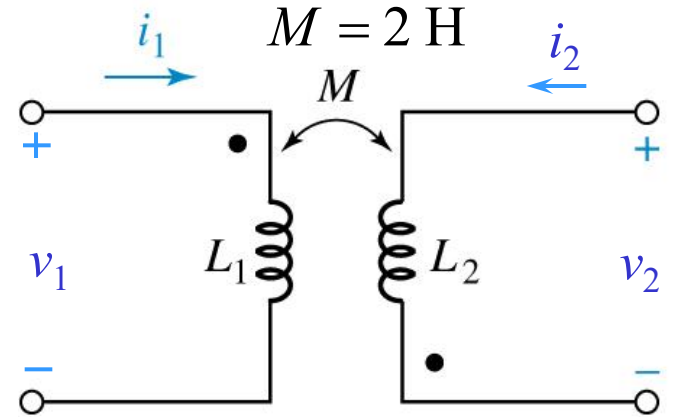
When two currents enter the dotted terminals on each coil, they produce the flux in the same direction.

Dot Convention

Example 13.1

(a) Find v_1 if $i_2 = 5\sin 45t$ A and $i_1 = 0$;

(b) Find v_2 if $i_1 = -8e^{-t}$ A and $i_2 = 0$.



$$(a) \quad v_1(t) = -M \frac{di_2}{dt}$$

$$= -2(45)(5 \cos 45t) = -450 \cos 45t \text{ V}$$

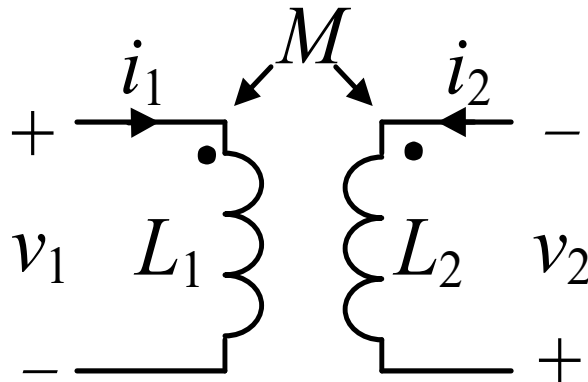
$$(b) \quad v_2(t) = -M \frac{di_1}{dt}$$

$$= -2(-1)(-8e^{-t}) = -16e^{-t} \text{ V}$$

Dot Convention

Example

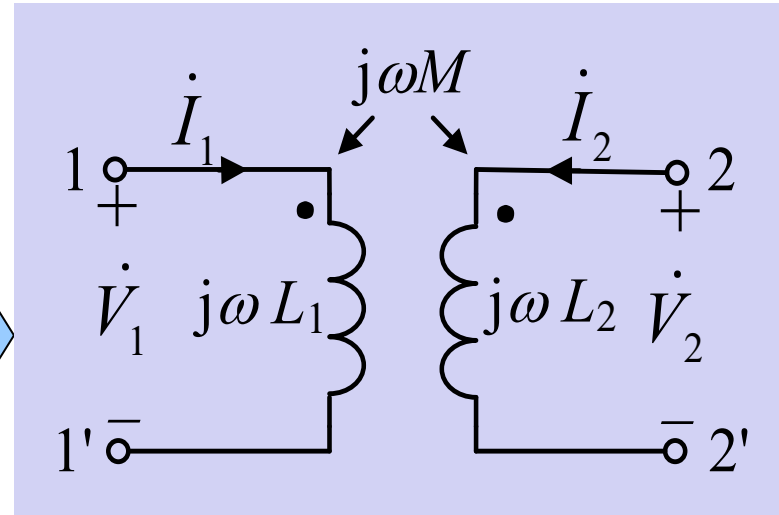
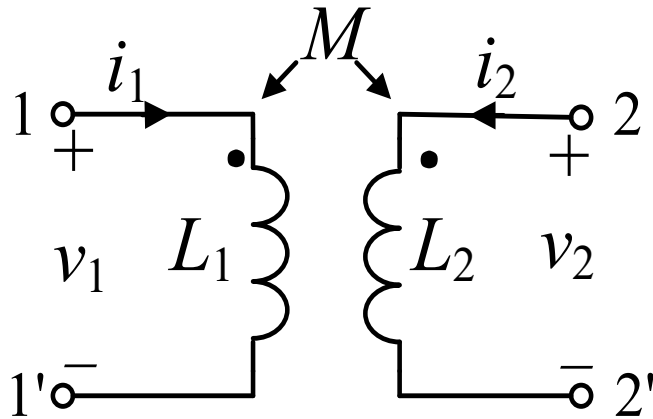
Express v_1 and v_2 with i_1 and i_2 .



$$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

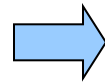
$$v_2 = -L_2 \frac{di_2}{dt} - M \frac{di_1}{dt}$$

Mutual Inductance in phasor Form



$$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$v_2 = L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$



$$\dot{V}_1 = j\omega L_1 \dot{I}_1 + j\omega M \dot{I}_2$$

$$\dot{V}_2 = j\omega M \dot{I}_1 + j\omega L_2 \dot{I}_2$$

Model of Mutual Inductance in phasor Form

Model of mutual inductance

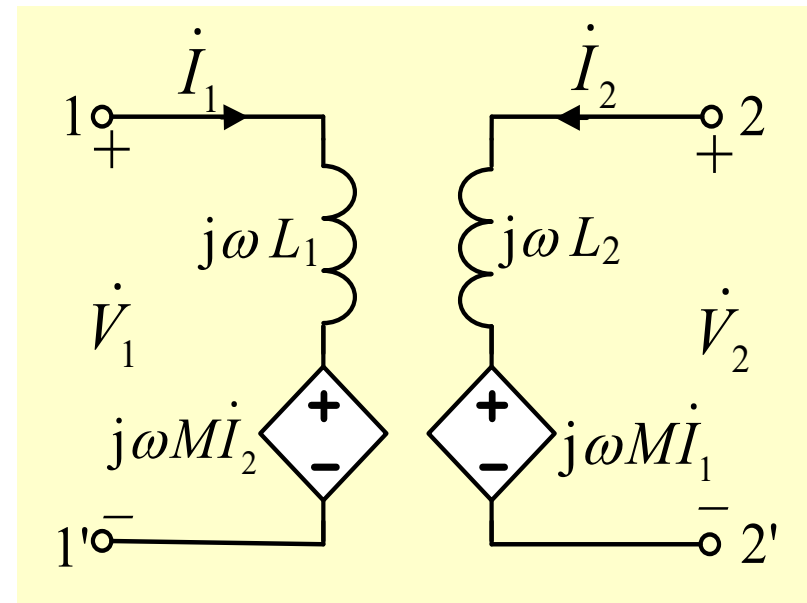
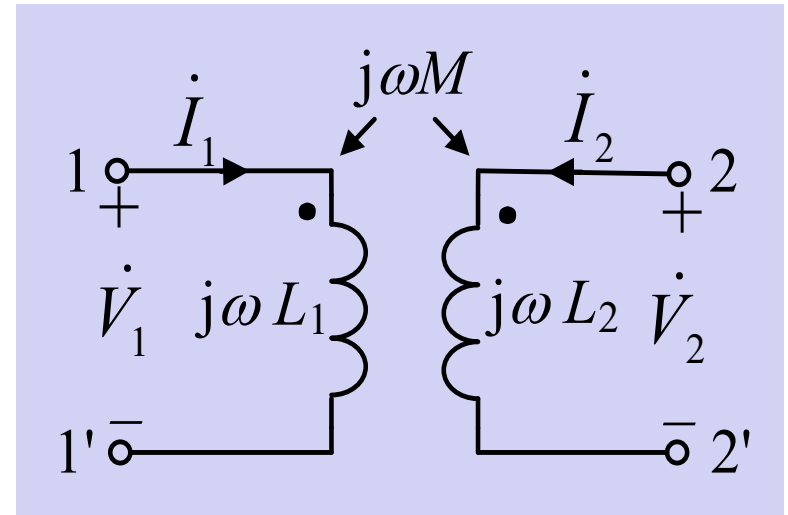
$$\dot{V}_1 = j\omega L_1 \dot{I}_1 + j\omega M \dot{I}_2$$

$$\dot{V}_2 = j\omega M \dot{I}_1 + j\omega L_2 \dot{I}_2$$



Fundamental analysis method:

Find the equivalent circuit

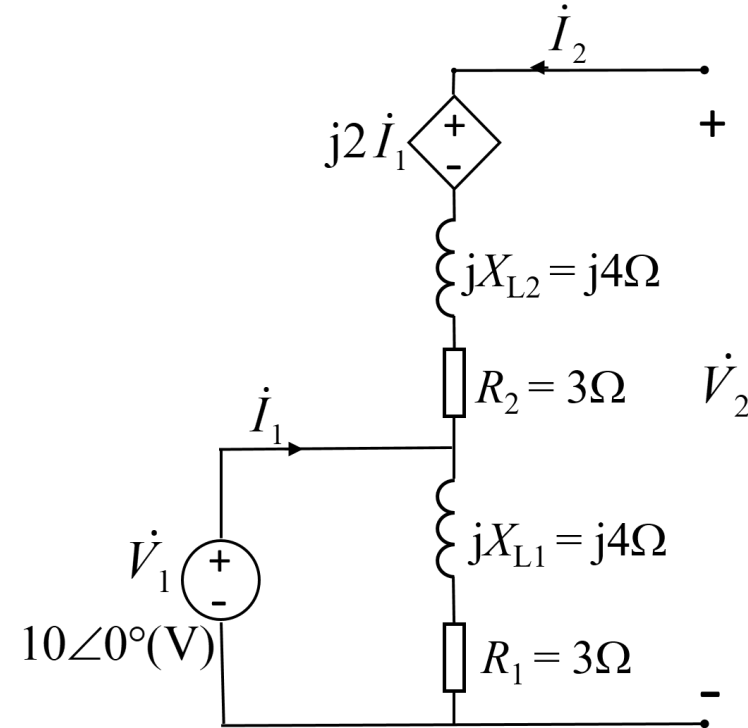
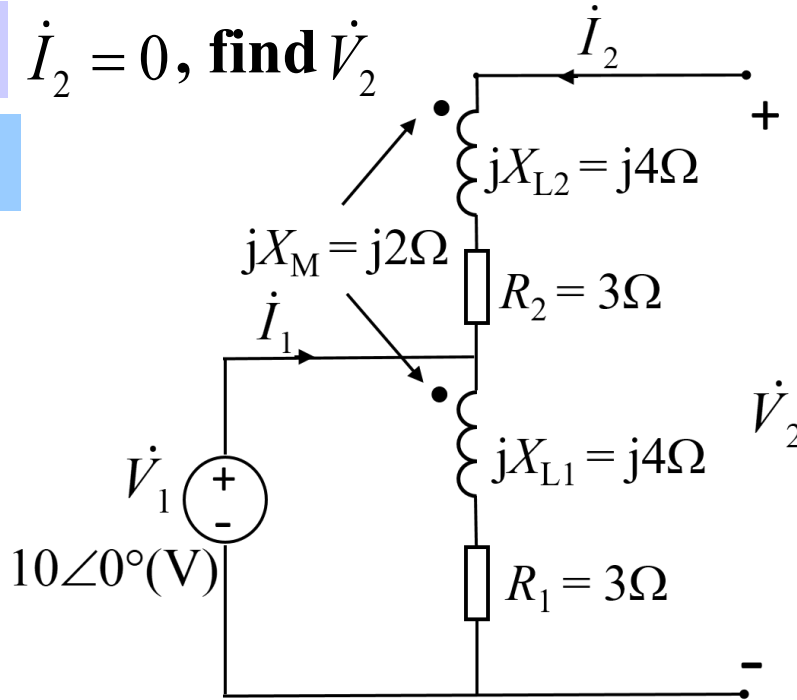


Model of Mutual Inductance in phasor Form

Example

$\dot{I}_2 = 0$, find $\dot{V}_2$

Solution



$$\dot{I}_1 = \frac{\dot{V}_1}{jX_{L1} + R_1} = \frac{10}{3 + j4} = 2 \angle -53.1^\circ (\text{A})$$

$$\begin{aligned} \dot{V}_2 &= \dot{V}_1 + j2\dot{I}_1 = 10 \angle 0^\circ + j2 \times 2 \angle -53.1^\circ \\ &= 13.4 \angle 10.3^\circ (\text{V}) \end{aligned}$$

Homework

- 1, 2, 3

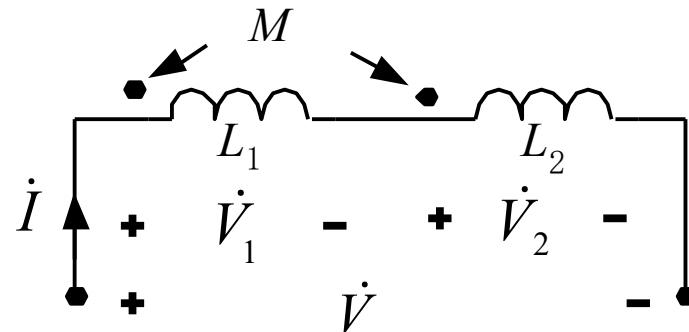
Connections of coupled inductors

Series connection

Series-aiding connection

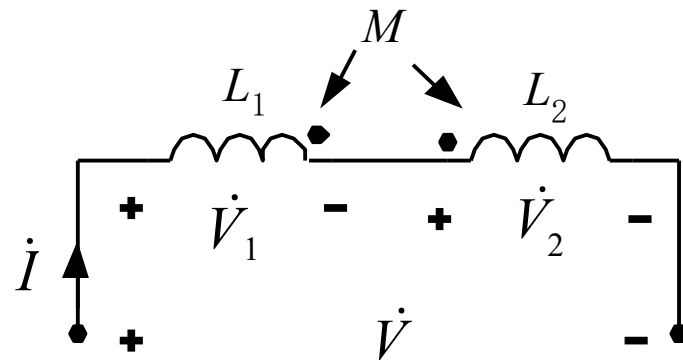
$$\begin{aligned}\dot{V} &= \dot{V}_1 + \dot{V}_2 = j\omega L_1 \dot{I} + j\omega M \dot{I} + j\omega L_2 \dot{I} + j\omega M \dot{I} \\ &= (L_1 + L_2 + 2M)j\omega \dot{I} = j\omega L_{\text{eq}} \dot{I}\end{aligned}$$

$$L_{\text{eq}} = L_1 + L_2 + 2M$$



Series-opposing connection

$$\begin{aligned}\dot{V} &= \dot{V}_1 + \dot{V}_2 = j\omega L_1 \dot{I} - j\omega M \dot{I} + j\omega L_2 \dot{I} - j\omega M \dot{I} \\ &= (L_1 + L_2 - 2M)j\omega \dot{I} = j\omega L_{\text{eq}} \dot{I}\end{aligned}$$



$$L_{\text{eq}} = L_1 + L_2 - 2M$$

$$L_1 + L_2 \geq 2M$$

$$M \leq \frac{L_1 + L_2}{2}$$

Connections of coupled inductors

Parallel connection

Same direction

$$L_{\text{eq}} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M}$$

reverse

$$L_{\text{eq}} = \frac{L_1 L_2 - M^2}{L_1 + L_2 + 2M}$$

$$L_{\text{eq}} \geq 0$$

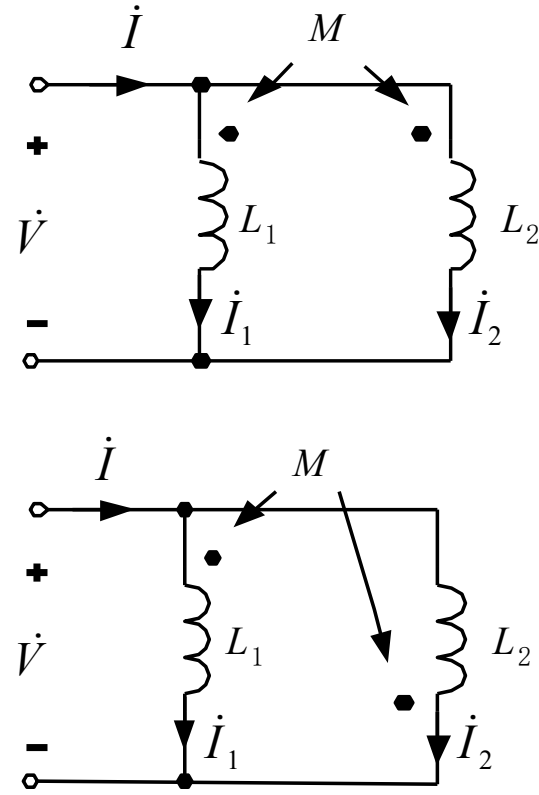
$$M \leq \sqrt{L_1 L_2}$$

$$\because \sqrt{L_1 L_2} \leq \frac{L_1 + L_2}{2}$$

$$\therefore M_{\text{max}} = \sqrt{L_1 L_2}$$

Coefficient of Coupling

$$k = \frac{M}{M_{\text{max}}} = \frac{M}{\sqrt{L_1 L_2}}$$

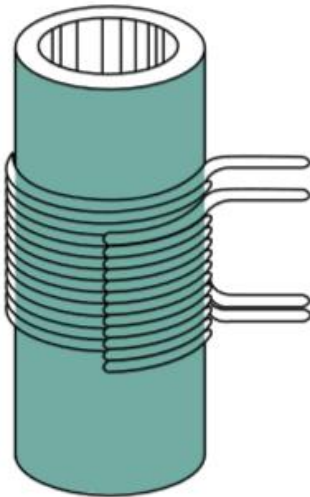


The Coupling Coefficient

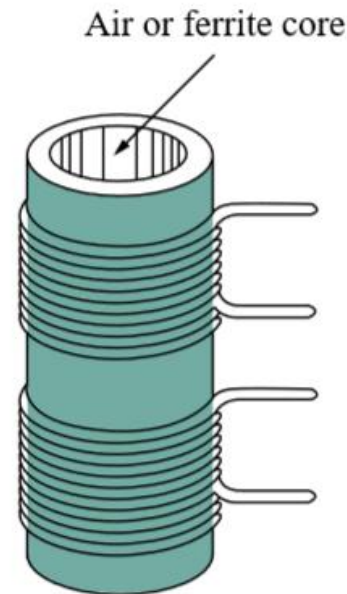
$$k = \frac{M}{M_{\max}} = \frac{M}{\sqrt{L_1 L_2}} \quad 0 \leq k \leq 1$$

The coupling coefficient is a measure of the magnetic coupling between two coils. It depends on the closeness of the two coils, their core, their orientation, and their windings

- $k = 1$: perfectly coupled



- $k > 0.5$: tightly coupled



- $k < 0.5$: loosely coupled

Energy Considerations

Demonstration of $M_{12} = M_{21}$ by Energy

Using $w(t) = \frac{1}{2} Li^2$, calculate the stored energy in coupled inductors at time t when $i_1(t) = I_1$, $i_2(t) = I_2$

Step1: $0 \rightarrow t_1 : i_1 = 0 \rightarrow I_1, i_2 = 0$

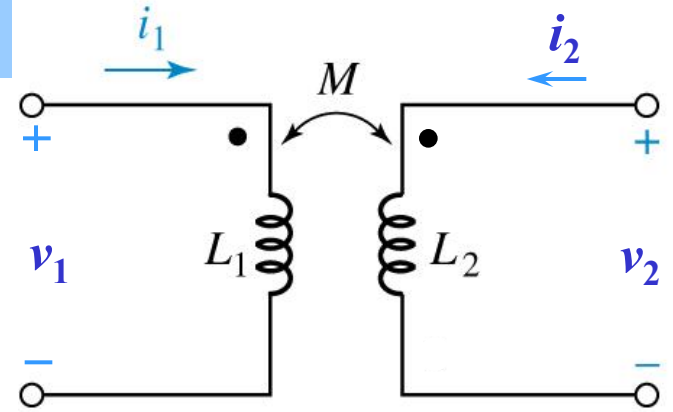
$$v_1 = L_1 \frac{di_1}{dt} \quad v_2 = M_{12} \frac{di_1}{dt} \quad p(t) = v_1 i_1 + v_2 i_2 = L_1 i_1 \frac{di_1}{dt}$$

$$w(t_1) = \int p(t) dt = L_1 \int_0^{I_1} i_1 di_1 = \frac{1}{2} L_1 I_1^2$$

Step2: $t_1 \rightarrow t : i_1 = I_1, i_2 = 0 \rightarrow I_2$

$$v_1 = L_1 \frac{dI_1}{dt} + M_{21} \frac{di_2}{dt} = M_{21} \frac{di_2}{dt} \quad v_2 = L_2 \frac{di_2}{dt}$$

$$p'(t) = v_1 I_1 + v_2 i_2 = M_{21} I_1 \frac{di_2}{dt} + L_2 i_2 \frac{di_2}{dt}$$



Energy Considerations

$$p(t) = v_1 i_1 + v_2 i_2 = L_1 i_1 \frac{di_1}{dt}$$

$$w(t_1) = \int p(t) dt = L_1 \int_0^{I_1} i_1 di_1 = \frac{1}{2} L_1 I_1^2$$

$$p'(t) = v_1 I_1 + v_2 i_2 = M_{21} I_1 \frac{di_2}{dt} + L_2 i_2 \frac{di_2}{dt}$$

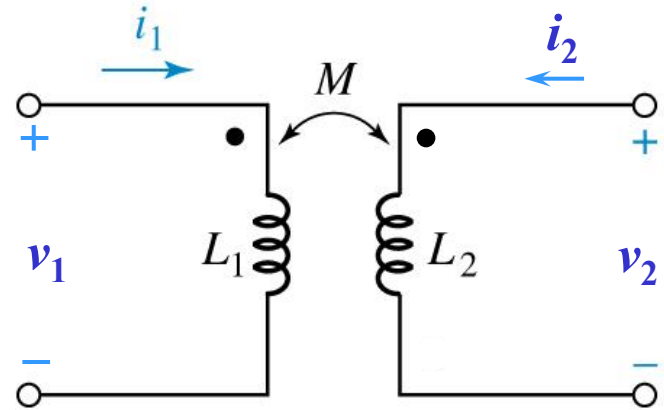
$$w'(t) = \int p'(t) dt = M_{21} I_1 \int_0^{I_2} di_2 + L_2 \int_0^{I_2} i_2 di_2 = M_{21} I_1 I_2 + \frac{1}{2} L_2 I_2^2$$

$$w(t) = w(t_1) + w'(t) = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 + M_{21} I_1 I_2$$

Now set step1, $0 \rightarrow t_1 : i_2 = 0 \rightarrow I_2, i_1 = 0$

step2, $t_1 \rightarrow t_2 : i_2 = I_2, i_1 = 0 \rightarrow I_1$

get $w = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 + M_{12} I_1 I_2$



If the initial and final conditions are the same, the stored energy must be equal.

$$M_{12} = M_{21} = M$$

Energy Considerations

For another dot convention case, get

$$w = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 - M I_1 I_2$$

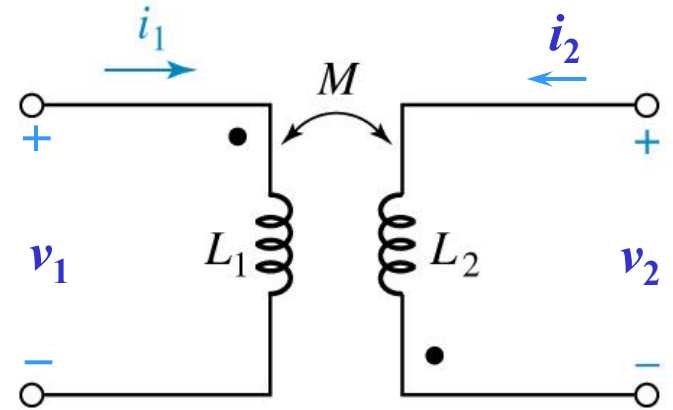
Conclusion

$$w(t) = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 \pm M i_1 i_2$$

Rewrite $w(t) = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 - M i_1 i_2$ in the form:

$$w(t) = \frac{1}{2} (\sqrt{L_1} i_1 - \sqrt{L_2} i_2)^2 + \sqrt{L_1 L_2} i_1 i_2 - M i_1 i_2$$

Since $w(t) \geq 0$, therefor $M \leq \sqrt{L_1 L_2}$

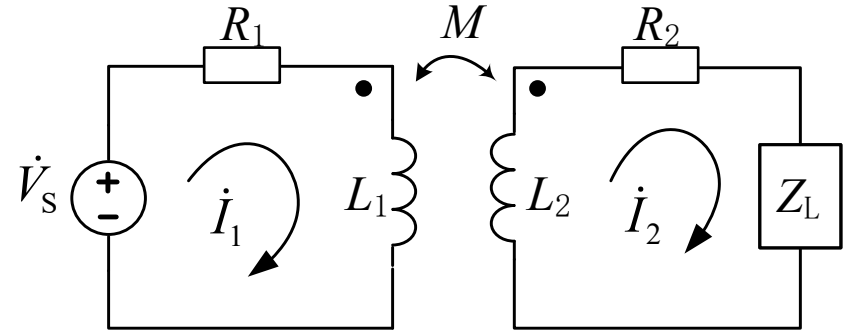


Analysis of Circuits with Mutual Inductance

The Linear Transformer

Linear Transformer

Transformer : a four-terminal device with two (or more) magnetically coupled coils.



Primary winding (coil, circuit), secondary winding

Linear transformer :

- ◆ coils are wound on a magnetically linear material (air, plastic, wood, etc.)
- ◆ sometimes called air-core transformer.
- ◆ $k \ll 1$
- ◆ Applications : radio frequency circuits.

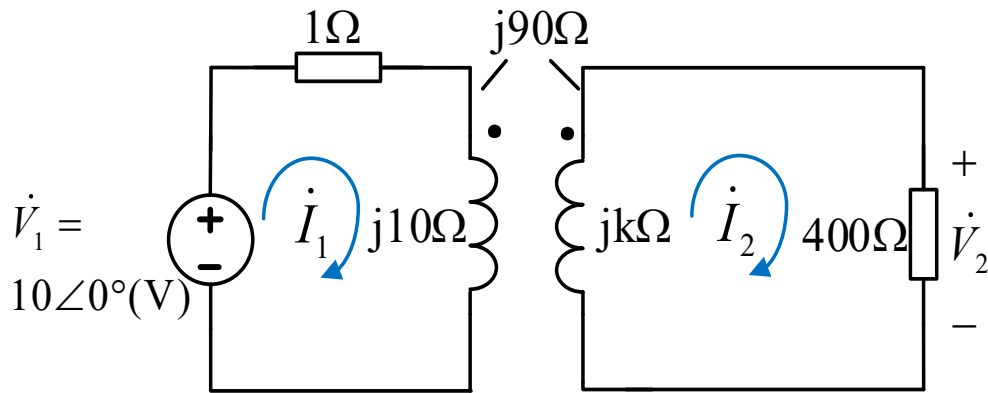
1. Systematic Mesh analysis

— loop analysis method

Example Find the phasor ratio of $\dot{V}_2 / \dot{V}_1$.

$$(1 + j10)\dot{I}_1 - j90\dot{I}_2 = 10$$

$$(400 + j1000)\dot{I}_2 - j90\dot{I}_1 = 0$$



Solve the equations for $\dot{I}_2$

$$\dot{I}_2 = 0.172\angle -16.7^\circ \text{ A}$$

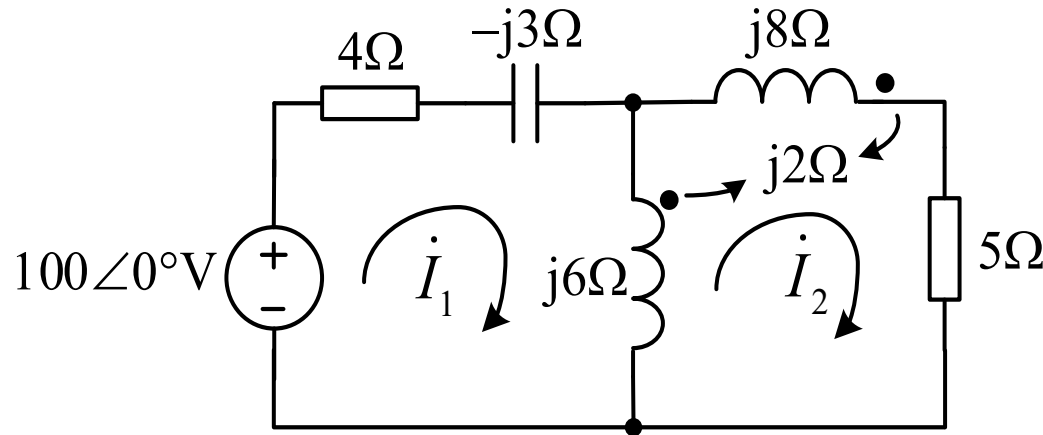
$$\frac{\dot{V}_2}{\dot{V}_1} = \frac{400(0.172\angle -16.7^\circ)}{10\angle 0^\circ} = 6.88\angle -16.7^\circ$$

1. Systematic Mesh analysis

— loop analysis method

Example

Calculate the mesh currents .

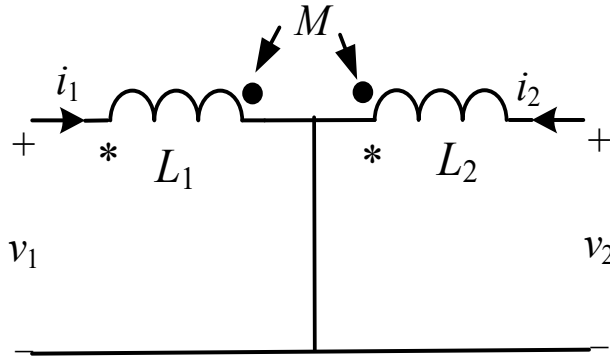


$$\begin{cases} (4 - j3 + j6)\dot{I}_1 - j6\dot{I}_2 - j2\dot{I}_2 = 100 \\ -j6\dot{I}_1 + (j6 + j8 + 5)\dot{I}_2 + j2\dot{I}_2 + j2(\dot{I}_2 - \dot{I}_1) = 0 \end{cases}$$

$$\begin{cases} (4 + j3)\dot{I}_1 - j8\dot{I}_2 = 100 \\ -j8\dot{I}_1 + (5 + j18)\dot{I}_2 = 0 \end{cases} \quad \begin{cases} \dot{I}_1 = \frac{\Delta_1}{\Delta} = \frac{100(5 + j18)}{30 + j87} = 20.3\angle 3.5^\circ \text{ A} \\ \dot{I}_2 = \frac{\Delta_2}{\Delta} = \frac{j800}{30 + j87} = 8.69\angle 19^\circ \text{ A} \end{cases}$$

2. Elimination of Mutual Inductance

Substitute T-shape circuit with a circuit without mutual inductance

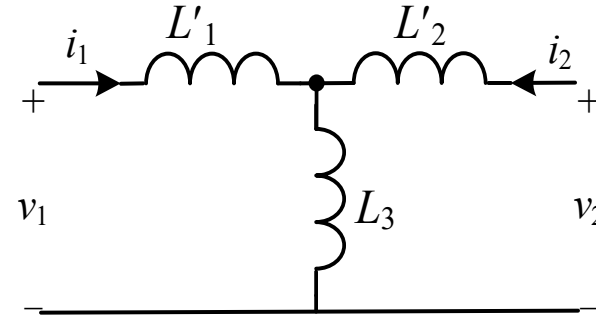


$$\dot{V}_1 = j\omega L_1 \dot{I}_1 \pm j\omega M \dot{I}_2$$

$$\dot{V}_2 = \pm j\omega M \dot{I}_1 + j\omega L_2 \dot{I}_2$$

Dotted terminals are connected

$$\begin{cases} L_3 = M \\ L_1' = L_1 - M \\ L_2' = L_2 - M \end{cases}$$



$$\dot{V}_1 = j\omega(L_1' + L_3) \dot{I}_1 + j\omega L_3' \dot{I}_2$$

$$\dot{V}_2 = j\omega L_3' \dot{I}_1 + j\omega(L_2' + L_3) \dot{I}_2$$

Dotted terminals are not connected

$$\begin{cases} L_3 = -M \\ L_1' = L_1 + M \\ L_2' = L_2 + M \end{cases}$$

Ex: Find i_1 and i_2 by Elimination of Mutual Inductance

$$R_1=R_2=1\Omega, L_1=2\text{H}, L_2=1\text{H}$$

$$k = 1/\sqrt{2} \quad C = 0.5\text{F}$$

$$v_s(t) = 2\sqrt{2} \cos t \text{ (V)}$$

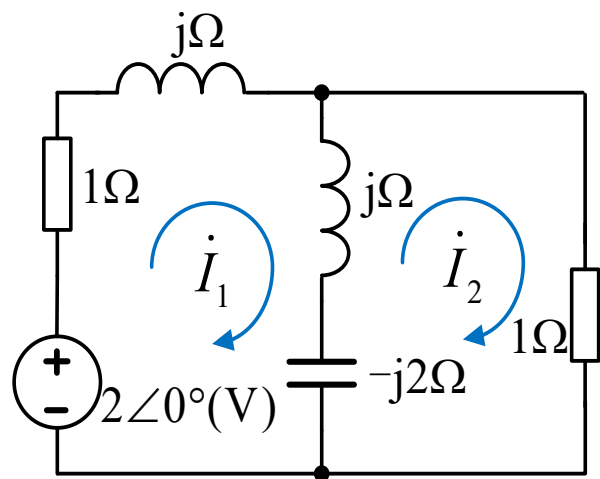
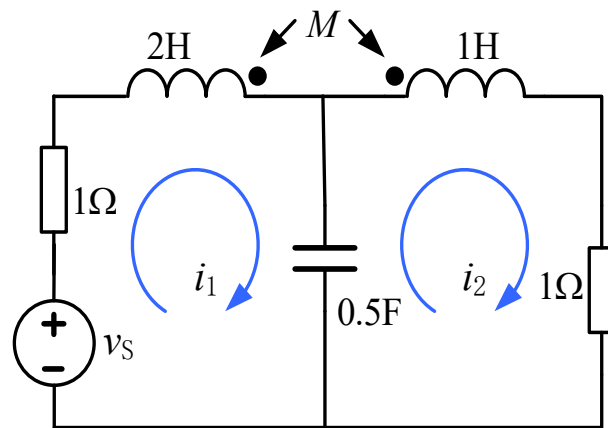
Solution

$$M = k\sqrt{L_1 L_2} = 1(\text{H})$$

After finding the equivalent circuit,
we have mesh equations:

$$\begin{cases} \dot{I}_{1\text{eff}} - (j - j2)\dot{I}_{2\text{eff}} = 2\angle 0^\circ \\ j\dot{I}_{1\text{eff}} + (1 - j)\dot{I}_{2\text{eff}} = 0 \end{cases}$$

$$\begin{cases} \dot{I}_{1\text{eff}} = \frac{6 - j2}{5} = 1.26\angle -18.4^\circ \text{ (A)} \\ \dot{I}_{2\text{eff}} = \frac{2 - j4}{5} = 0.89\angle -63.4^\circ \text{ (A)} \end{cases}$$



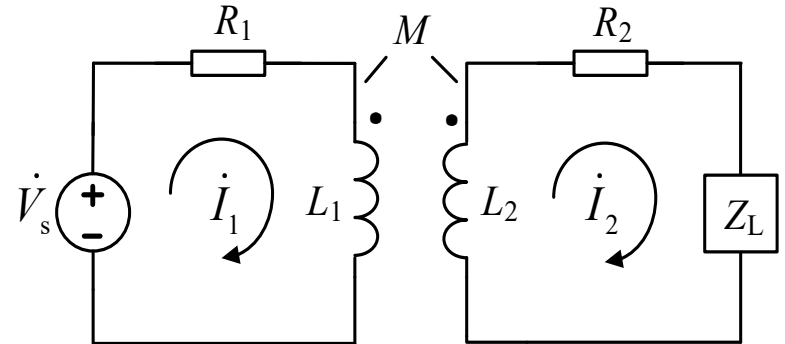
$$i_1(t) = 1.26\sqrt{2} \cos(t - 18.4^\circ) \text{ (A)}$$

$$i_2(t) = 0.89\sqrt{2} \cos(t - 63.4^\circ) \text{ (A)}$$

3. Analysis by Reflected Impedance

Reflected Impedance

$$\begin{cases} (R_1 + j\omega L_1)\dot{I}_1 - j\omega M\dot{I}_2 = \dot{V}_s \\ -j\omega M\dot{I}_1 + (R_2 + j\omega L_2 + Z_L)\dot{I}_2 = 0 \end{cases}$$



Define

$$\begin{aligned} Z_{11} &= R_1 + j\omega L_1 \\ Z_{22} &= R_2 + j\omega L_2 + Z_L \end{aligned} \quad \begin{cases} Z_{11}\dot{I}_1 - j\omega M\dot{I}_2 = \dot{V}_s \\ -j\omega M\dot{I}_1 + Z_{22}\dot{I}_2 = 0 \end{cases}$$

$$Z_{\text{in}} = \frac{\dot{V}_s}{\dot{I}_1} = Z_{11} + \frac{(\omega M)^2}{Z_{22}} = Z_{11} + Z_{1r}$$

Reflected impedance

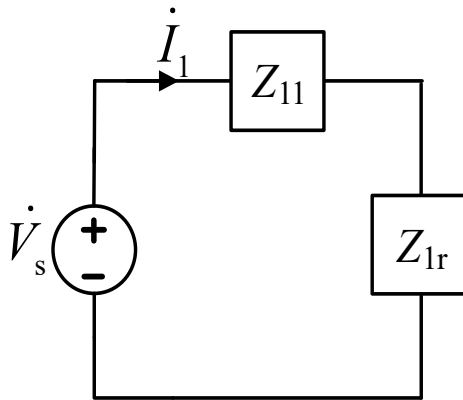
$$Z_{1r} = \frac{(\omega M)^2}{Z_{22}}$$

3. Analysis by Reflected Impedance

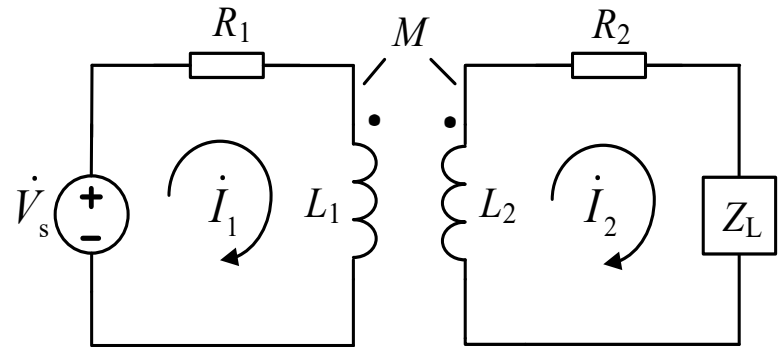
Input Impedance

$$Z_{\text{in}} = \frac{\dot{V}_s}{\dot{I}_1} = Z_{11} + \frac{(\omega M)^2}{Z_{22}}$$

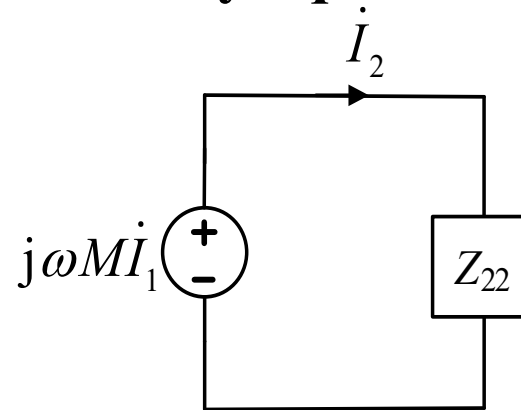
• Primary equivalent circuit:



$$\dot{I}_1 = \frac{\dot{V}_s}{Z_{11} + \frac{(\omega M)^2}{Z_{22}}} = \frac{\dot{V}_s}{Z_{11} + Z_{1r}}$$



• Secondary equivalent circuit:



$$\dot{I}_2 = \frac{j\omega M\dot{I}_1}{Z_{22}}$$

Power on Reflected Impedance

Reflected Resistance and Reflected Reactance

$$\begin{aligned} Z_{1r} &= \frac{(\omega M)^2}{Z_{22}} = \frac{(\omega M)^2}{Z_{22} Z_{22}^*} Z_{22}^* \quad (Z_{22} = R_{22} + jX_{22}) \\ &= \frac{\omega^2 M^2}{R_{22}^2 + X_{22}^2} \cdot R_{22} + j \frac{\omega^2 M^2}{R_{22}^2 + X_{22}^2} (-X_{22}) = R_{1r} + jX_{1r} \end{aligned}$$

R_{1r} : Reflected Resistance

X_{1r} : Reflected Reactance, $\rightarrow$ opposite reactance to X_{22}

$$P_{1r} = I_1^2 R_{1r} = I_1^2 \frac{\omega^2 M^2}{R_{22}^2 + X_{22}^2} \cdot R_{22} = \frac{(\omega M I_1)^2}{R_{22}^2 + X_{22}^2} \cdot R_{22} = I_2^2 R_{22} = P_2$$

The power dissipation of secondary equals to that of R_{1r} .

Ex $X_M = 1\Omega$, Find I_1, I_2 , the source's supplying power and absorbing power in secondary loop.

$$Z_{1r} = \frac{\omega^2 M^2}{Z_{22}} = \frac{1}{1+j} = \frac{1}{2} - j\frac{1}{2} \Omega$$

$$\dot{I}_1 = \frac{\dot{V}_s}{Z_{11} + Z_{1r}} = \frac{10}{j2 + \frac{1}{2} - j\frac{1}{2}} = 2 - j6 \text{ A}$$

$\dot{V}_s$ **Supply power**

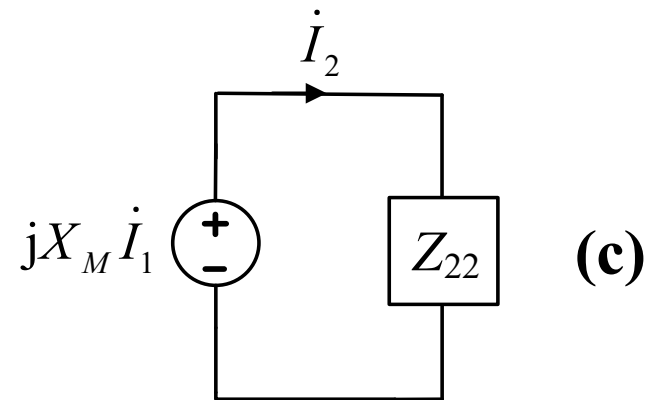
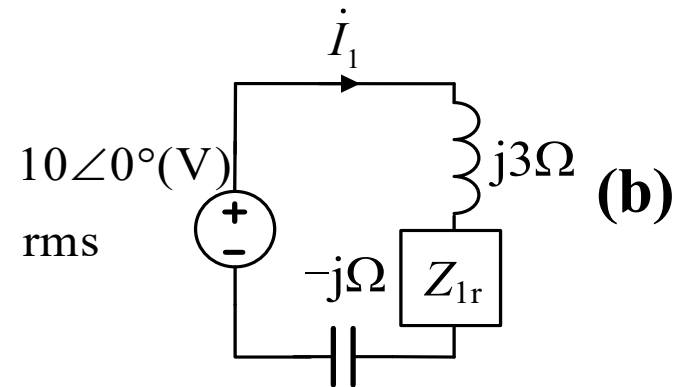
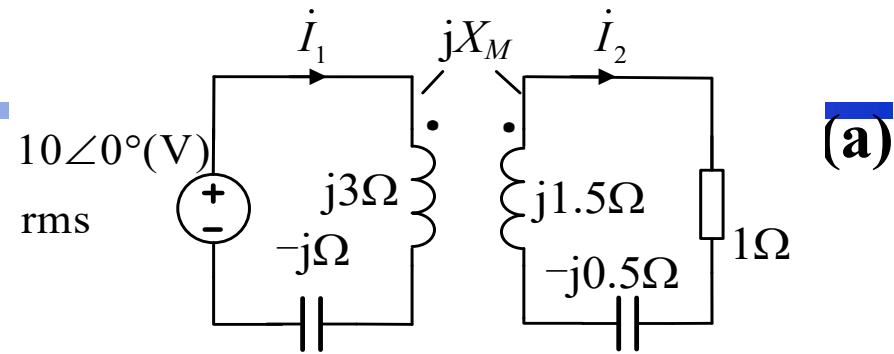
$$P_s = \text{Re}[\dot{V}_s \dot{I}_1^*] = \text{Re}[10(2 + j6)] = 20(\text{W})$$

Absorbing power in secondary loop

$$\dot{I}_2 = \frac{j\omega M \dot{I}_1}{Z_{22}} = \frac{j1 \times (2 - j6)}{1 + j1.5 - j0.5} = 4 - j2(\text{A})$$

$$P_R = I_2^2 R_{22} = (4^2 + 2^2) \times 1 = 20(\text{W})$$

or
$$P_R = I_1^2 R_{1r} = (2^2 + 6^2) \times \frac{1}{2} = 20(\text{W})$$



Ex

$X_M = 1\Omega$, Find I_1, I_2 by
Thevenin's theory and
absorbing P of 2nd loop.

$$\dot{V}_{oc} = j\omega M \frac{\dot{V}_s}{Z_{11}} = j \frac{10}{j3 - j} = 5(V)$$

$$Z_{2r} = \frac{\omega^2 M^2}{Z_{11}} = \frac{1^2}{j3 - j} = -j0.5(\Omega)$$

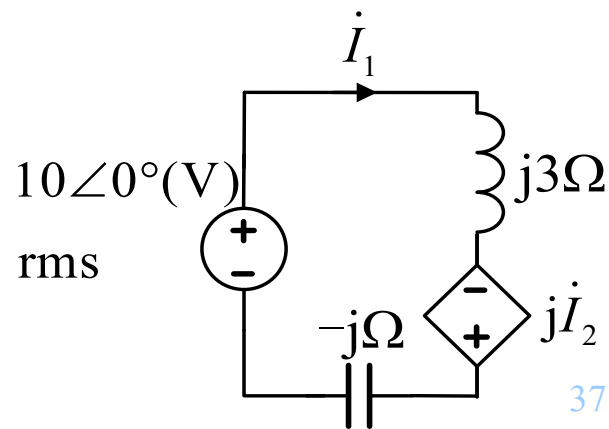
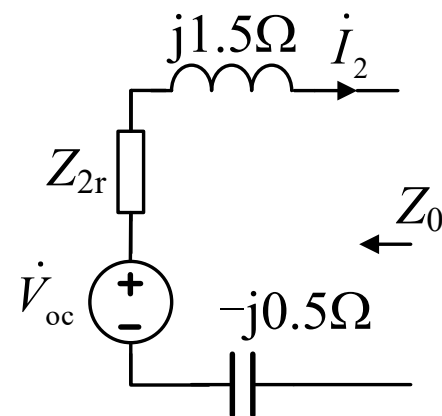
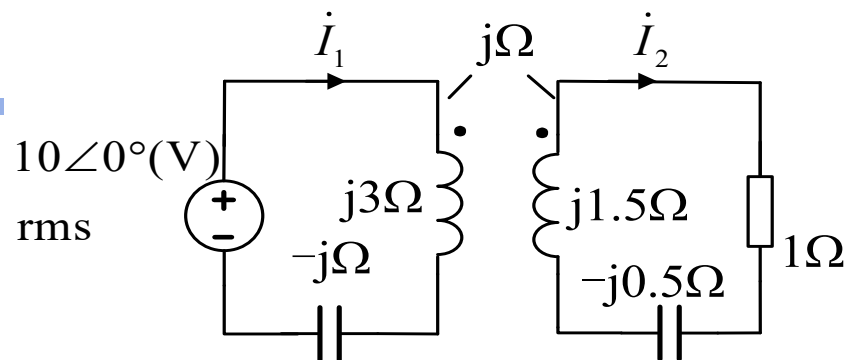
$$Z_0 = Z_{2r} + j1.5 - j0.5$$

$$\dot{I}_2 = \frac{\dot{V}_{oc}}{Z_0 + 1} = \frac{5}{-j0.5 + j1.5 - j0.5 + 1} = 4 - j2(A)$$

$$P_2 = I_2^2 R_{22} = (4^2 + 2^2) \times 1 = 20(W)$$

$$\dot{I}_1 = \frac{\dot{V}_s + j\omega M \dot{I}_2}{Z_{11}} = \frac{10 + j \times (4 - j2)}{j3 - j}$$

$$= 2 - j6(A)$$



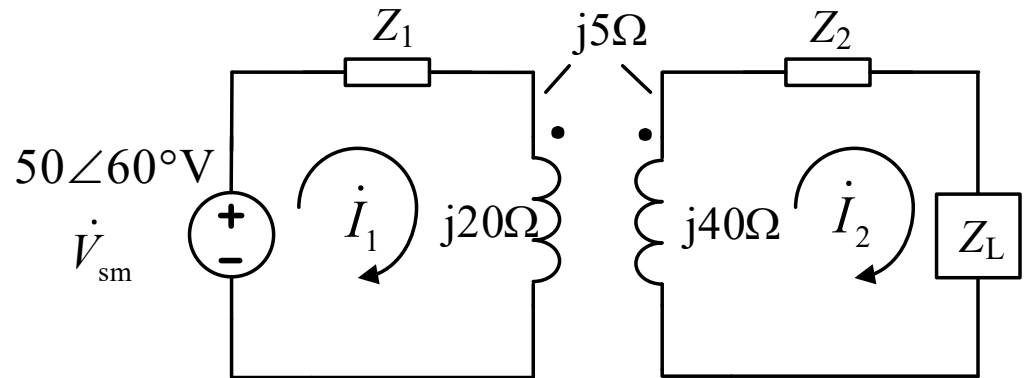
Example

Calculate the input impedance, I_1 and power of Z_L ,

Given $Z_1 = 60 - j100\Omega$,

$$Z_2 = 30 + j40\Omega,$$

$$Z_L = 80 + j60\Omega$$

Solution

$$Z_{in} = Z_1 + j20 + Z_{lr}$$

$$= 60 - j100 + j20 + \frac{5^2}{30 + j40 + j40 + 80 + j60} = 100\angle -53.1^\circ \Omega$$

$$\dot{I}_1 = \frac{\dot{V}_s}{Z_{in}} = \frac{50\angle 60^\circ}{100\angle -53.1^\circ} = 0.5\angle 113.1^\circ \text{ A}$$

$$\dot{I}_2 = \frac{j5\dot{I}_1}{j40 + Z_2 + Z_L} = \frac{2.5\angle 203.1^\circ}{110 + j140} = 0.014\angle 151.3^\circ \text{ A}$$

$$P_L = \frac{1}{2} I_2^2 \text{Re}[Z_L] = \frac{1}{2} (0.014)^2 (80) = 7.84 \text{ mW}$$

Example

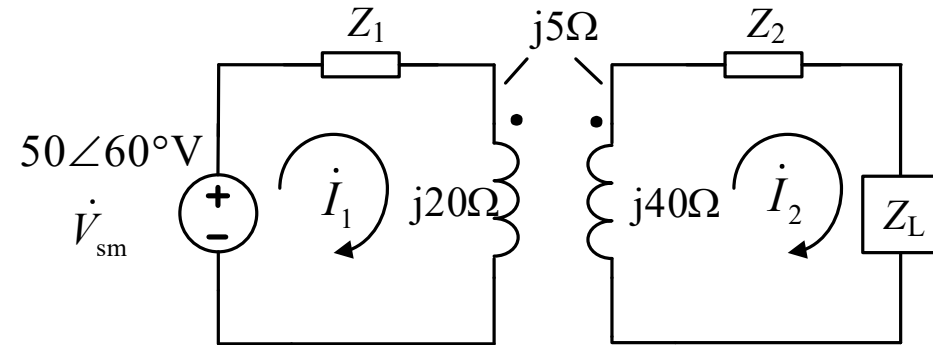
Calculate the input impedance, I_1 and power of Z_L ,

Given $Z_1 = 60 - j100\Omega$,

$$Z_2 = 30 + j40\Omega,$$

$$Z_L = 80 + j60\Omega$$

Solution



$$Z_{in} = Z_1 + j20 + Z_{lr}$$
$$= 60 - j100 + j20 + \frac{5^2}{30 + j40 + j40 + 80 + j60} = 100\angle -53.1^\circ \Omega$$

$$\dot{I}_1 = \frac{\dot{V}_s}{Z_{in}} = \frac{50\angle 60^\circ}{100\angle -53.1^\circ} = 0.5\angle 113.1^\circ \text{ A}$$

$$\dot{I}_2 = \frac{j5\dot{I}_1}{j40 + Z_2 + Z_L} = \frac{2.5\angle 203.1^\circ}{110 + j140} = 0.014\angle 151.3^\circ \text{ A}$$

$$P_L = \frac{1}{2} I_2^2 \text{Re}[Z_L] = \frac{1}{2} (0.014)^2 (80) = 7.84 \text{ mW}$$

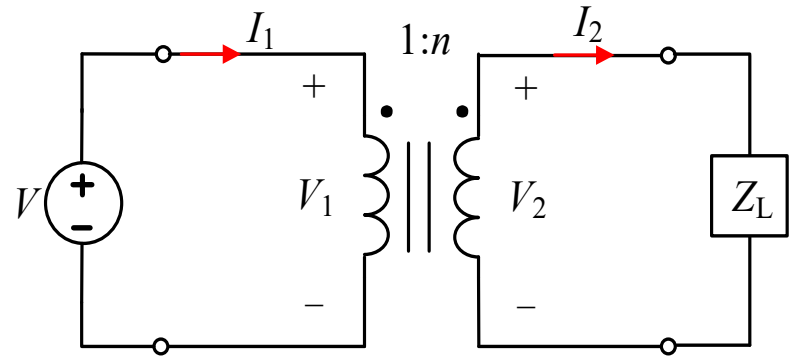
Homework

- 4, 5, 7, 9, 10

Ideal Transformers

Ideal Transformers

Ideal Transformer



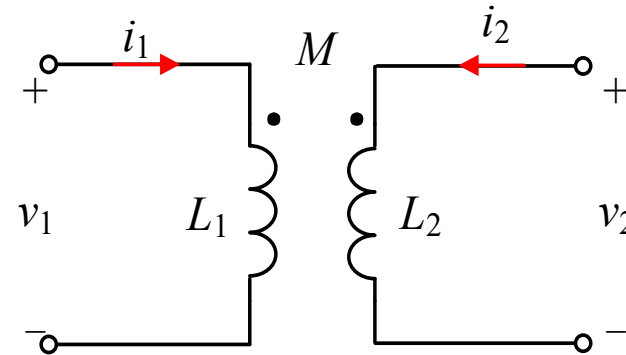
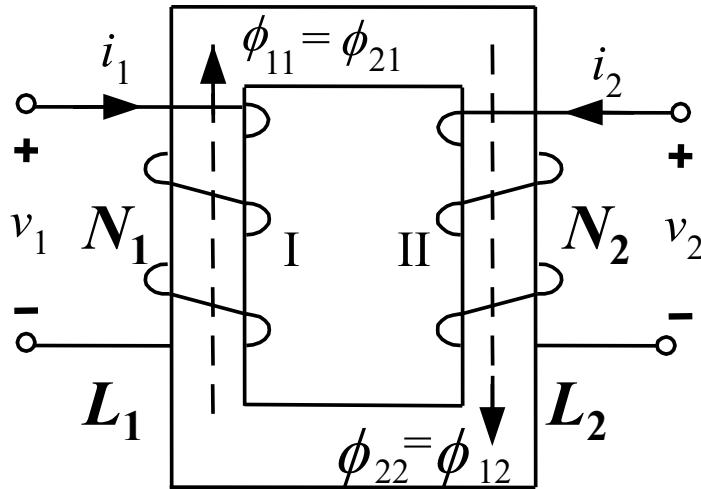
- ◆ Coils have very large reactance ($L_1, L_2, M \rightarrow \infty$)
- ◆ Coupling coefficient is equal to unity ($k = 1$).
- ◆ Primary and secondary coils are lossless ($R_1 = 0 = R_2$).

Real Transformer

- ◆ Well-designed iron-core transformer,
- ◆ tightly coupled, very large reactance of coils,
- ◆ very small wire resistance.

Ideal transformer and coupled inductance

coils $\xrightarrow{\text{No loss}}$ coupled inductance $\xrightarrow{k=1}$ Perfectly coupled transformer $\xrightarrow{L_1, L_2 \rightarrow \infty}$ ideal transformer



Perfectly Coupled transformer ($v_1 \sim v_2$)

L_1 and L_2 are not infinite, with $k=1$

$$\because k=1 \quad M = \sqrt{L_1 L_2}$$

$$\dot{V}_1 = j\omega L_1 \dot{I}_1 + j\omega M \dot{I}_2$$

$$\dot{V}_2 = j\omega M \dot{I}_1 + j\omega L_2 \dot{I}_2$$

$$\dot{I}_1 = (\dot{V}_1 - j\omega M \dot{I}_2) / j\omega L_1$$

$$\dot{V}_2 = j\omega L_2 \dot{I}_2 + \frac{M \dot{V}_1}{L_1} - \frac{j\omega M^2 \dot{I}_2}{L_1}$$

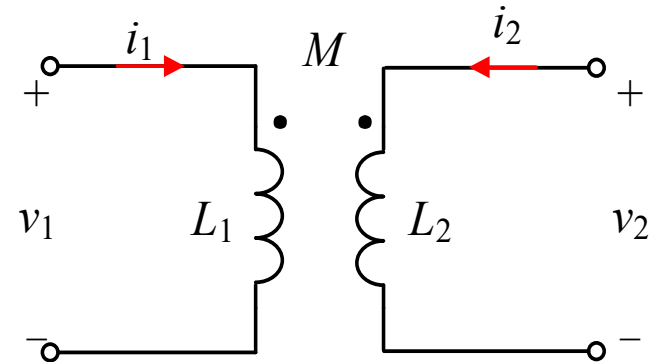
$$= j\omega L_2 \dot{I}_2 + \frac{\sqrt{L_1 L_2} \dot{V}_1}{L_1} - \frac{j\omega L_1 L_2 \dot{I}_2}{L_1} = \sqrt{\frac{L_2}{L_1}} \dot{V}_1 = n \dot{V}_1$$

Conclusion:

$$\dot{V}_2 = n \dot{V}_1$$

$$v_2 = n v_1$$

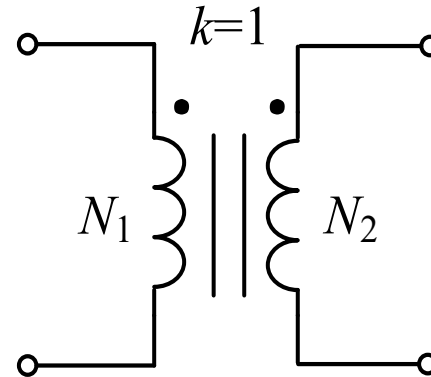
$$n = \sqrt{\frac{L_2}{L_1}}$$



Perfectly Coupled transformer

Turns Ratio

$$\psi = L i$$



$$\phi \propto Ni$$

$$\psi \propto N\phi$$

$$L \propto N^2$$

The self-inductance of a coil is proportional to N^2 .

$$\frac{L_2}{L_1} = \frac{N_2^2}{N_1^2} = n^2$$

Turns ratio:

$$n = \frac{N_2}{N_1} = \sqrt{\frac{L_2}{L_1}}$$

$n > 1$: step-up transformer

$n < 1$: step-down transformer

Perfectly Coupled transformer ($i_1 \sim i_2$)

$$k=1 \quad M = \sqrt{L_1 L_2} \quad n = \sqrt{\frac{L_2}{L_1}} \quad v_2 = n v_1$$

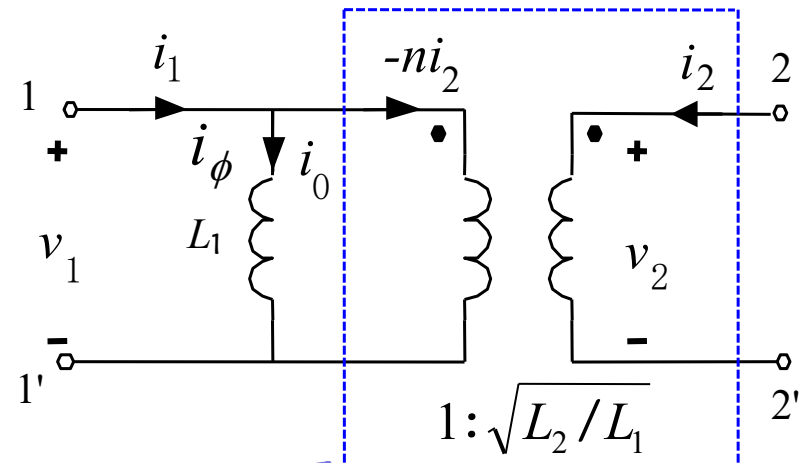
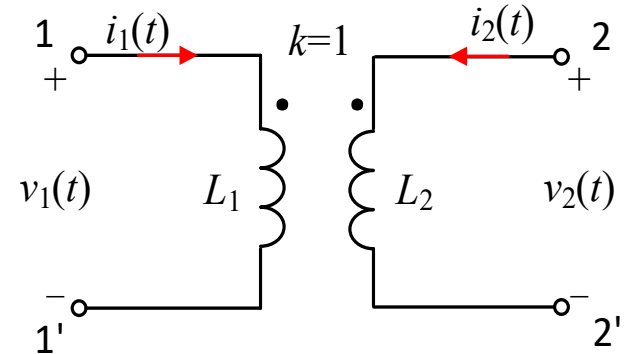
$$v_1 = L_1 \frac{di_1}{dt} + \sqrt{L_1 L_2} \times \frac{di_2}{dt}$$

$$\frac{di_1}{dt} = \frac{v_1}{L_1} - \sqrt{\frac{L_2}{L_1}} \times \frac{di_2}{dt}$$

integral

$$i_1 = \frac{1}{L_1} \int_{-\infty}^t v_1(\tau) d\tau - n i_2 = i_\phi + i_1'$$

where $i_1' = -n i_2$

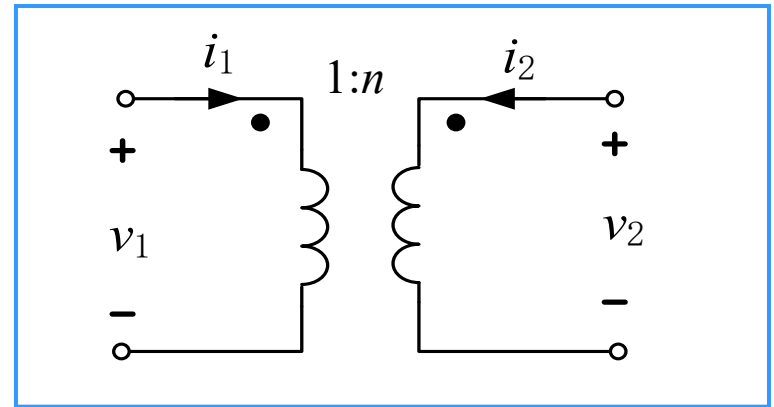


Ideal transformer

Ideal Transformers

Voltage relationship

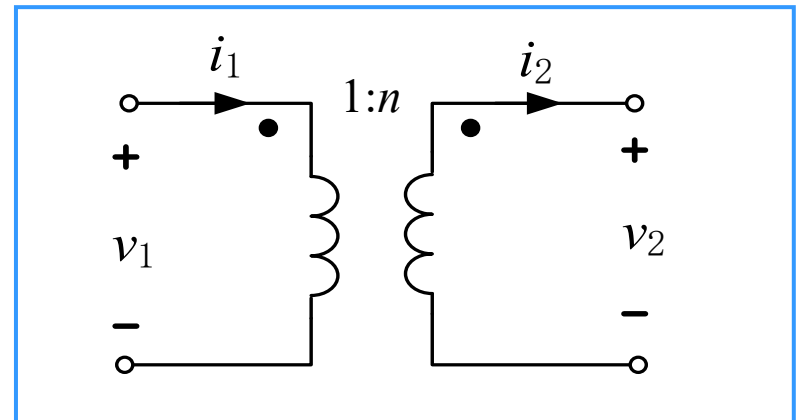
$$\frac{v_2}{v_1} = n$$



Current relationship

$$i_1 = -ni_2$$

$$i_1 = ni_2$$



Ideal Transformers

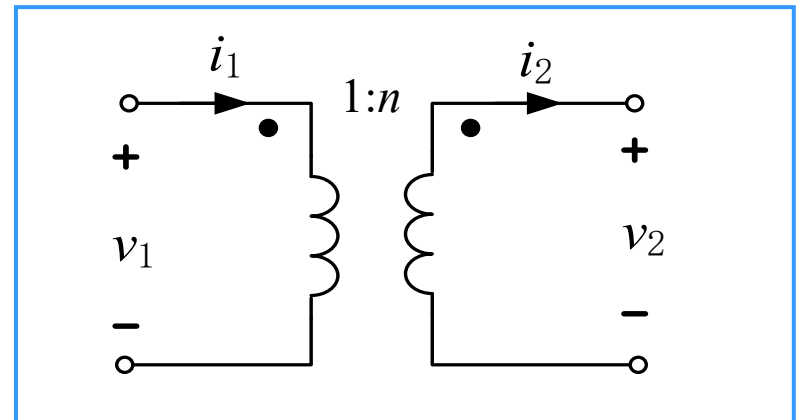
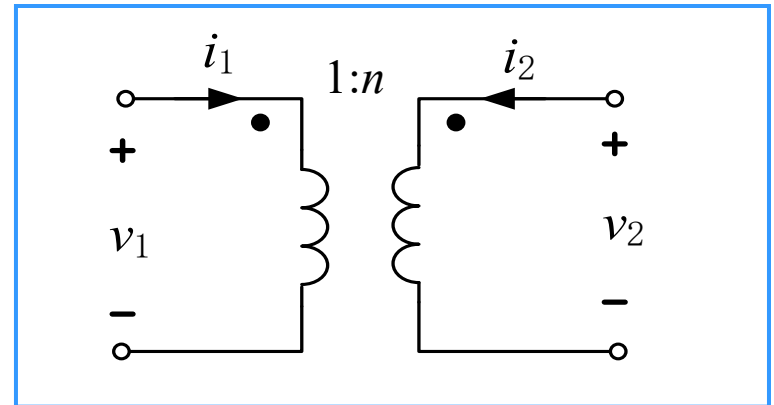
Lossless Element

Use rms phasors for sinusoidal steady-state circuit :

$$\begin{aligned}\tilde{S} &= \tilde{S}_1 + \tilde{S}_2 = \dot{V}_1 \dot{I}_1^* + \dot{V}_2 \dot{I}_2^* \\ &= \dot{V}_1 \dot{I}_1^* + n \dot{V}_1 \left(-\frac{1}{n} \dot{I}_1^* \right) = 0\end{aligned}$$

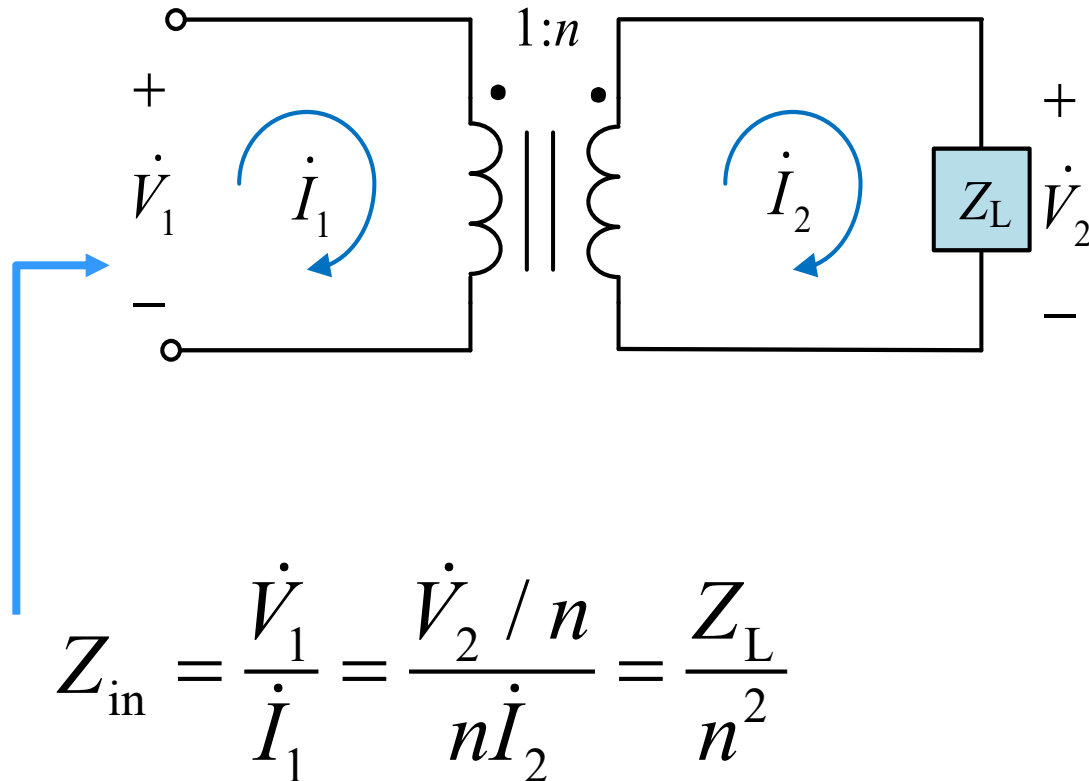
Instantaneous power

$$\begin{aligned}p(t) &= p_1(t) + p_2(t) = v_1 i_1 + v_2 i_2 \\ &= v_1 (-n i_2) + n v_1 i_2 = 0\end{aligned}$$



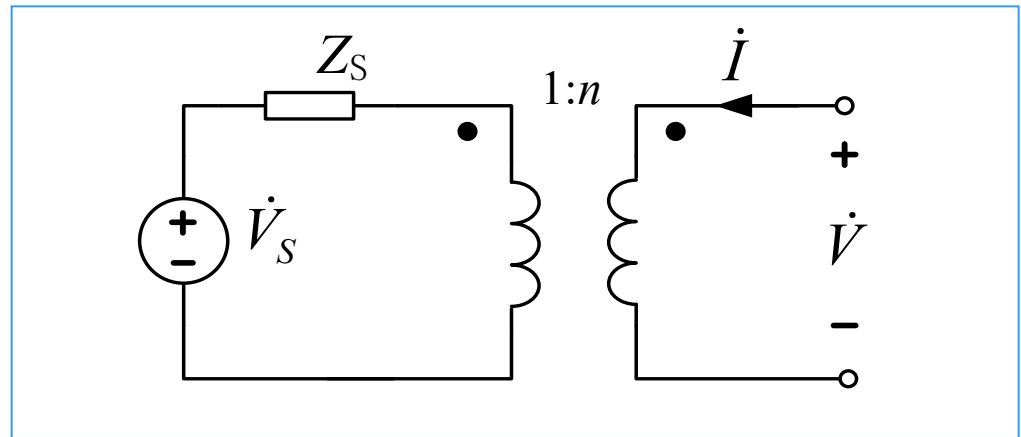
Ideal Transformers

Reflected Impedance



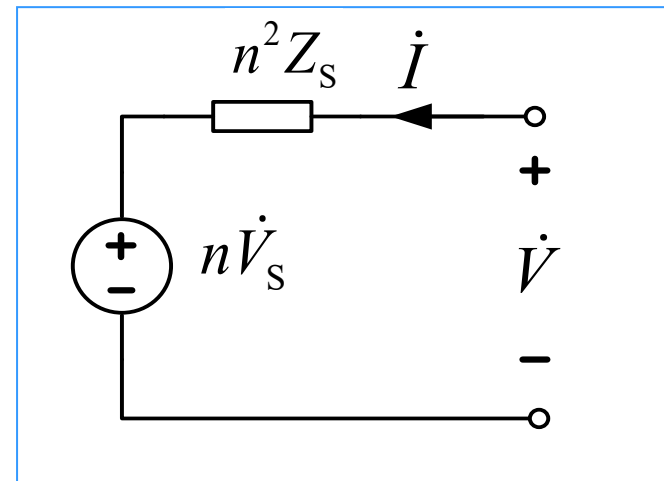
Ideal Transformers

Equivalent Circuits



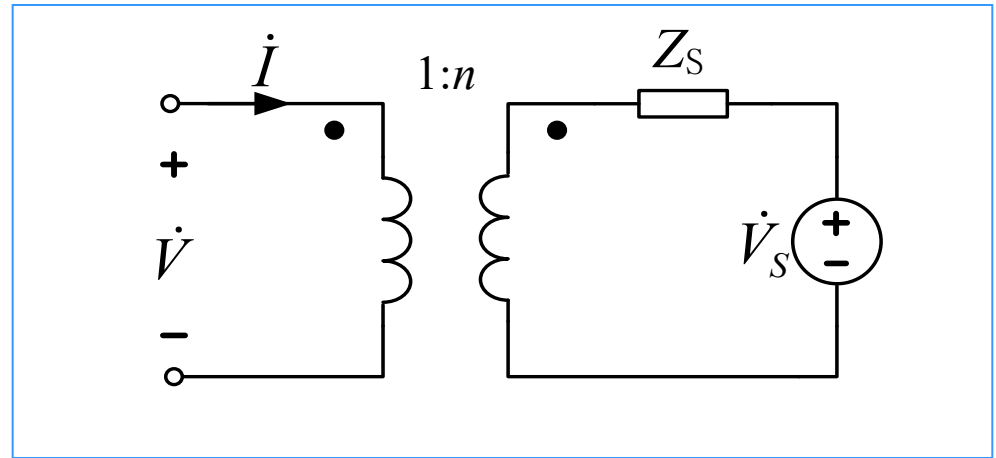
$$\dot{V}_{OC} = n\dot{V}_S$$

$$Z_0 = n^2 Z_S$$



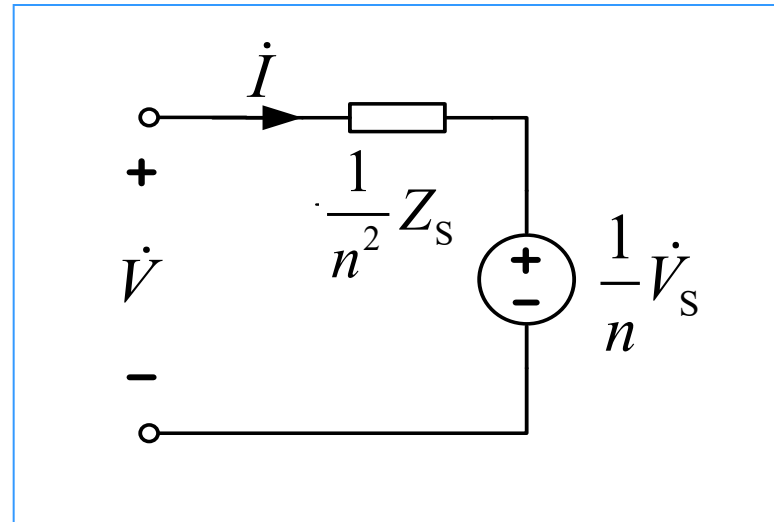
Ideal Transformers

Equivalent Circuits



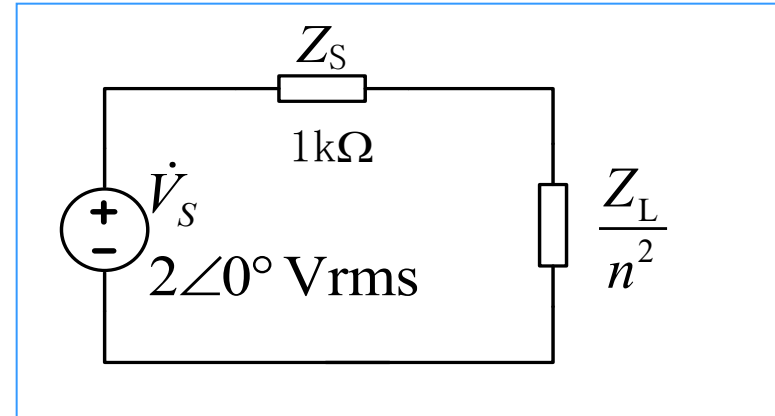
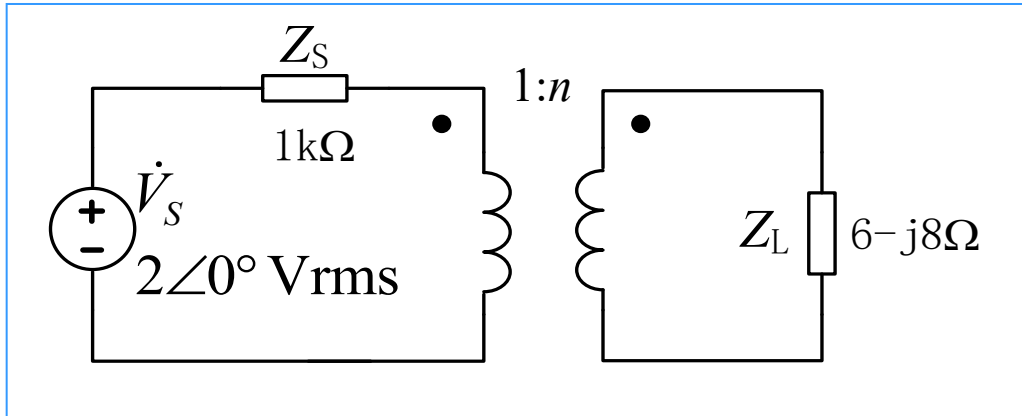
$$\dot{V}_{\text{OC}} = \frac{1}{n} \dot{V}_S$$

$$Z_0 = \frac{1}{n^2} Z_S$$



Example

Determine the turn ratio of the matching transformer so that load Z_L can draw maximum average power.



When $Z_s = |Z_L| / n^2$ Z_L gets max power.

$$n = \sqrt{\frac{|Z_L|}{R}} = \sqrt{\frac{10}{1000}} = 0.1 \quad Z_{\text{in}} = \frac{Z_L}{n^2} = 100(6 - j8) \Omega$$

$$\dot{I}_1 = \frac{\dot{V}_s}{R_s + Z_{\text{in}}} = \frac{2\angle 0^\circ}{1600 - j800} = 1.1\angle 26.6^\circ \text{ mA}$$

$$P_{L\text{max}} = I_1^2 \text{Re}[Z_{\text{in}}] = 1.1^2 \times 10^{-6} \times 600 = 0.726 \text{ mW}$$

Example

Find the power absorbed by 10Ω resistor. 120V is the maximal amplitude of the voltage source.

Mesh equation:

$$(20 + 30)\dot{I}_1 - 30\dot{I}_2 + \dot{V}_1 = 120$$

$$-30\dot{I}_1 + (10 + 30)\dot{I}_2 - \dot{V}_2 = 0$$

$$\dot{V}_2 = -\frac{1}{2}\dot{V}_1$$

$$\dot{I}_2 = -2\dot{I}_1$$

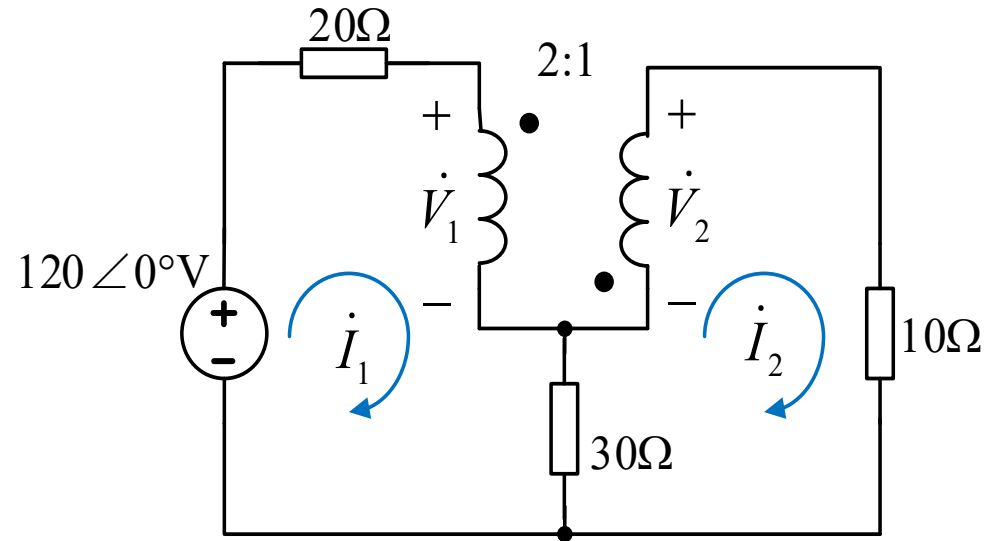
Substitute into mesh equations,

$$-55\dot{I}_2 - 2\dot{V}_2 = 120$$

$$15\dot{I}_2 + 40\dot{I}_2 - \dot{V}_2 = 0$$

$$\dot{I}_2 = -120/165 = -0.727$$

$$P = \frac{1}{2} \times 0.727^2 (10) = 2.65(\text{W})$$



Autotransformer

Example Find the input impedance Z_i .

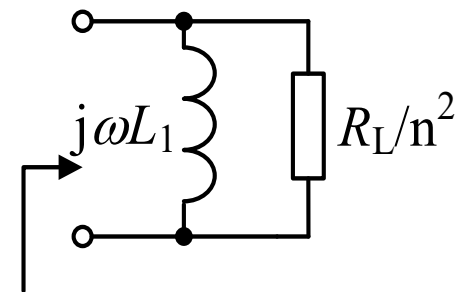
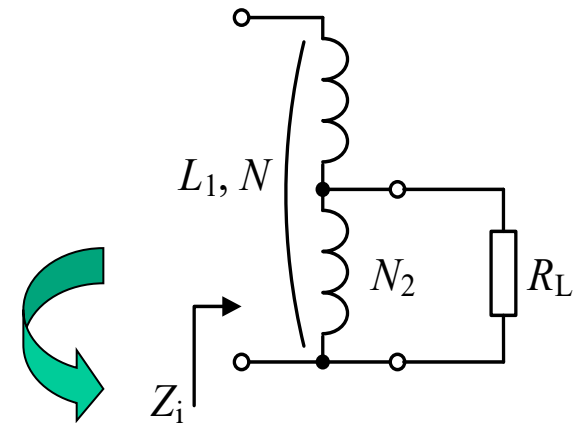
Given $L_1 = 0.574\text{H}$ $N_2 = N / 3$
 $R_L = 40\ \Omega$ $\omega = 314\text{ rad/s}$

Solution For autotransformer, $k=1$

$$n = \frac{N_2}{N_1} = \frac{1}{3} \quad \frac{R_L}{n^2} = 360\ \Omega$$

$$Z_i = j\omega_0 L_1 // \frac{R_L}{n^2}$$

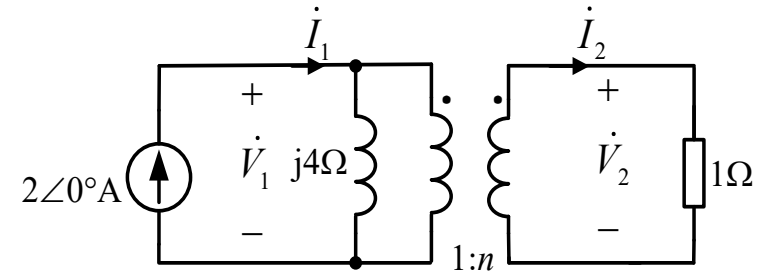
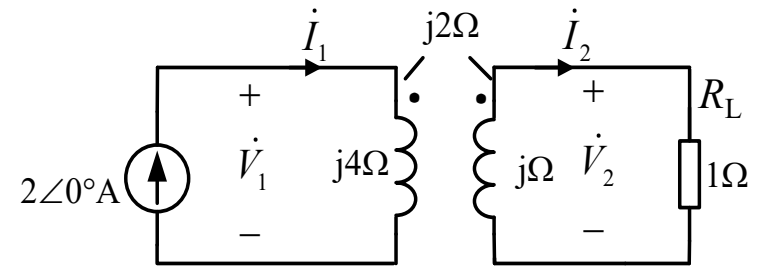
$$Z_i = \frac{j180 \times 360}{j180 + 360} = 161 \angle 63.4^\circ\ \Omega$$



Example**Find $\dot{I}_1$, $\dot{I}_2$, $\dot{V}_1$ and $\dot{V}_2$** **Solution****(1) Find k**

$$k = \frac{M}{\sqrt{L_1 L_2}} = \frac{\omega M}{\sqrt{X_{L1} X_{L2}}} = \frac{2}{\sqrt{4}} = 1$$

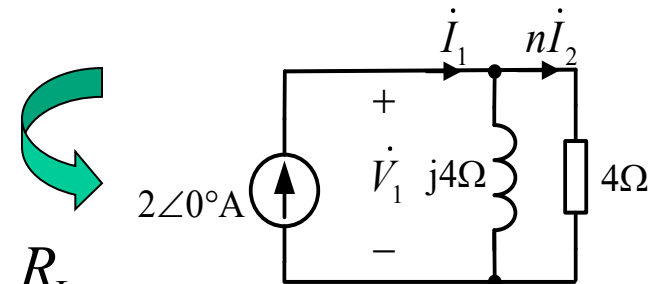
$$n = \sqrt{\frac{L_2}{L_1}} = \sqrt{\frac{X_{L2}}{X_{L1}}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

(2) Draw the equivalent circuit with ideal transformer**(3) Draw the primary equivalent circuit**

$$\frac{R_L}{n^2} = 4\Omega \quad \dot{I}_1 = \dot{I}_s = 2\angle 0^\circ \text{ A}$$

$$n\dot{I}_2 = \dot{I}_1 \frac{j4}{4 + j4} = \sqrt{2}\angle 45^\circ \text{ A}$$

$$\dot{I}_2 = 2\sqrt{2}\angle 45^\circ \text{ A}$$



$$\dot{V}_1 = n\dot{I}_2 \frac{R_L}{n^2}$$

$$= \sqrt{2}\angle 45^\circ \cdot 4 = 4\sqrt{2}\angle 45^\circ \text{ V}$$

$$\dot{V}_2 = n\dot{V}_1 = 2\sqrt{2}\angle 45^\circ \text{ V}$$

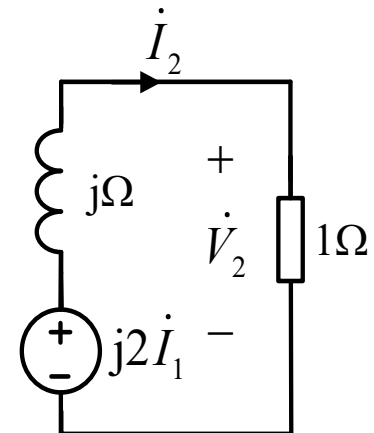
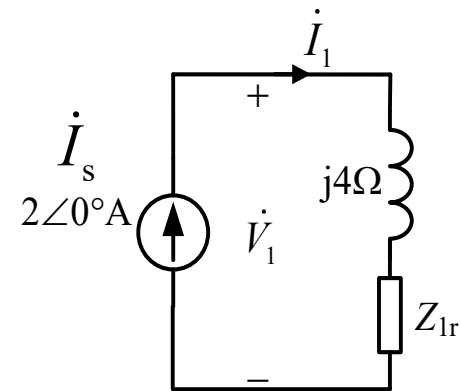
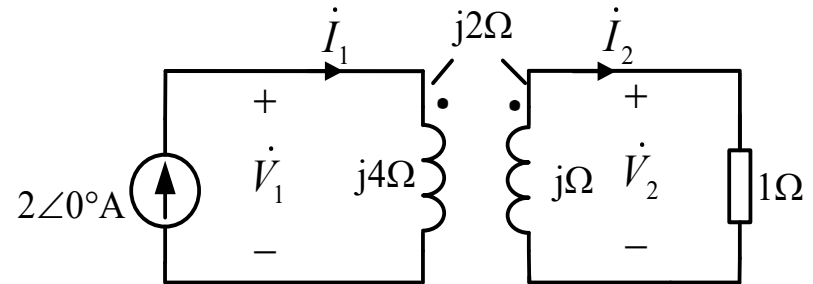
Alternative method: Reflected Impedance

$$Z_{lr} = \frac{2^2}{1+j} = 2 - j2 \quad \Omega$$

$$\begin{aligned}\dot{V}_1 &= \dot{I}_s (j4 + 2 - j2) \\ &= 4\sqrt{2} \angle 45^\circ \text{ (V)}\end{aligned}$$

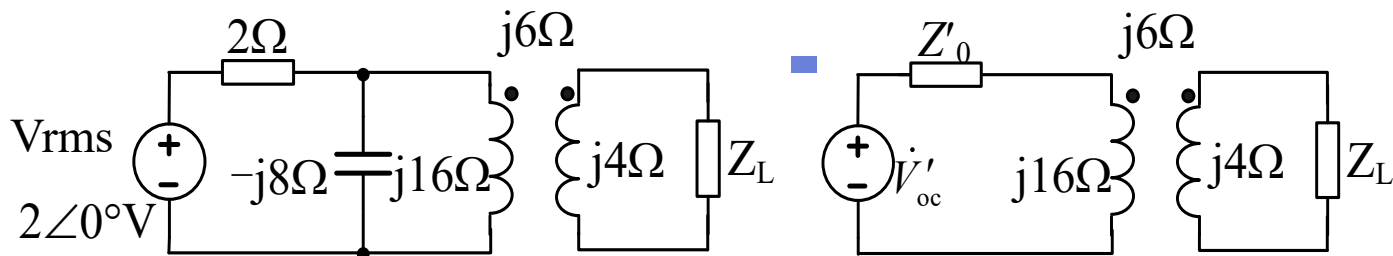
$$\begin{aligned}\dot{I}_2 &= \frac{j2\dot{I}_s}{1+j} = 2 + j2 \\ &= 2\sqrt{2} \angle 45^\circ \text{ (A)}\end{aligned}$$

$$\dot{V}_2 = \dot{I}_2 R_L = 2\sqrt{2} \angle 45^\circ \text{ (V)}$$



Example

Find $Z_L = ?$ $P_{L\max}$
 $P_{L\max} = ?$



Solution Find Thevenin equivalence

$$\dot{V}'_{oc} = \frac{-j8}{2 - j8} \times 2 = 1.94 \angle -14^\circ \quad (\text{V})$$

$$Z'_0 = \frac{2 \times (-j8)}{2 - j8} = 1.9 \angle -14^\circ \quad (\Omega)$$

$$\dot{V}_{oc} = j6 \times \frac{\dot{V}'_{oc}}{Z'_0 + j16} = 0.75 \angle -7.1^\circ \quad (\text{V})$$

$$\text{When } Z_L = 0.277 - j1.71 \quad (\Omega)$$

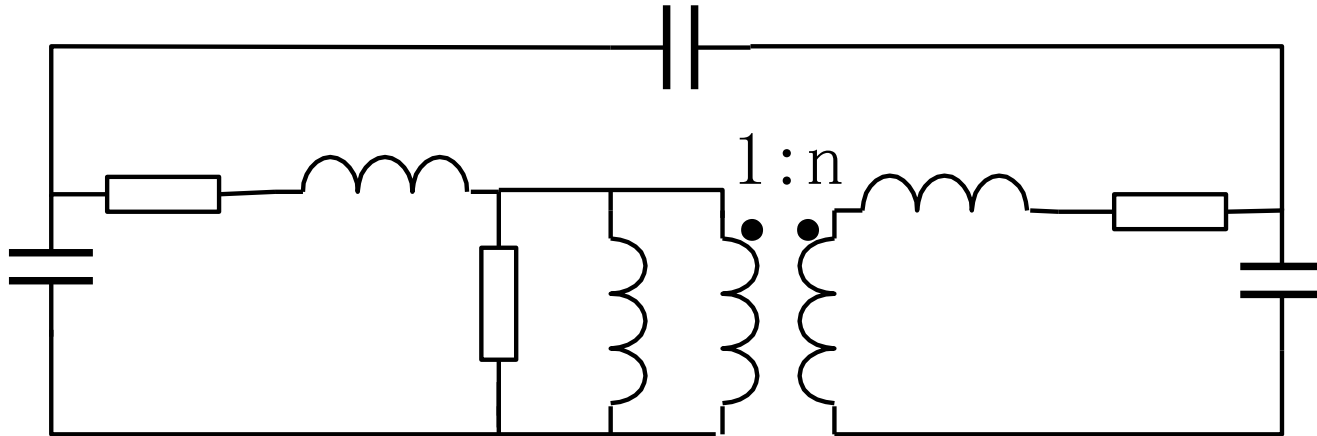
$$Z_0 = j4 + \frac{6^2}{Z'_0 + j16} = 0.277 + j1.71 (\Omega) \quad P_{L\max} = \frac{0.75^2}{4 \times 0.277} = 0.5 (\text{W})$$

$$\begin{aligned}
 V_{oc} &= j6 \times \frac{1.94 \angle -14^\circ}{j16 + 1.94 \angle -14^\circ} \\
 &= j6 \times \frac{1.94 \angle -14^\circ}{j16 + 1.94(0.97 - 0.24j)} \\
 &= j6 \times \frac{1.94 \angle -14^\circ}{1.882 + j15.527} \\
 &= 6 \angle 90^\circ \times \frac{1.94 \angle -14^\circ}{15.64 \angle 83.09^\circ} \\
 &= 0.75 \angle -7.1^\circ (\text{V})
 \end{aligned}$$

Real transformer

If loss is not 0, there is R .

If it works at high frequency, there is C .



Homework

- **5.12, 5.13, 5.17, 5.19**